Analysis of Household Surveys

Session notes

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September 10, 2026

Contents

1 Design, Sampling, and Collection

1.1 Reading

- Deaton ch. 1 — the whole chapter
- Deaton §1.4 on survey design and §2.2 on what to do about it
- Skim the GLSS7 *Report* methodology section

Deaton is open access (CC BY 3.0 IGO). On the hub it is already in your `reading/` folder; otherwise PDF.

The GLSS7 report is published by the Ghana Statistical Service and is not ours to redistribute: GSS catalogue 97.

1.2 Setup

```
%xmode Plain

# These notebooks are read two ways: on your own screen, and projected onto
# a wall in a lit room where thin lines and pale markers are simply not
# there.  Those want different plots, so the difference is a switch rather
# than a compromise that suits neither.  Leave it False; the lecturer flips
# it.  Everything below the 'if' is about the wall, not about your laptop.
PROJECTOR = False

import matplotlib as mpl
mpl.rcParams.update({'figure.dpi': 110, 'font.size': 11,
                     'lines.linewidth': 1.8})
if PROJECTOR:
    mpl.rcParams.update({
        'figure.figsize': (7, 4.5),
        'font.size': 15, 'axes.labelsize': 15, 'axes.titlesize': 17,
        'xtick.labelsize': 13, 'ytick.labelsize': 13, 'legend.fontsize': 13
        'lines.linewidth': 3.0, 'lines.markersize': 9, 'axes.linewidth': 1.
    })

import lsms_library as ll
import pandas as pd
```

1.3 Brief description of LSMS Surveys

A timeline of the surveys `lsms_library` knows about: one row per country, one bar per survey wave, so we can see overall temporal coverage — 37 countries, 113 waves, 1985 to the present.

1.4 Topical coverage of LSMS Surveys

You can check current coverage yourself:

```
cov = ll.coverage()
cov
```

`sane` means the library builds that feature for that wave and the result passes its checks; `absent` and `undeclared` mean it is not there to build.

Summarizing:

```
print("countries: %d | features: %d | waves: %d"
      % (cov.country.nunique(), cov.feature.nunique(),
         cov[cov.wave != ""].wave.nunique()))
cov.tier.value_counts()
```

So what is available for a country you care about:

```
cov.head(10)
```

1.5 LSMS_Library is a work in progress

Open source; see `https://github.com/ligon/lsms_library`. The idea is to provide a software *abstraction layer* to interface with LSMS-style surveys, providing a set of common labels, tables, and variables.

Note that this is **not** "harmonization"; we preserve all the detail of the individual surveys. Aggregation across surveys where necessary is left to the analyst.

If you find a problem, something's missing, or have a feature request: `https://github.com/ligon/lsms_library/issues`. I'd really appreciate your help!

1.6 Explore Example Country

What countries are available?

```
ll.countries()
```

I'll use Uganda as my example; it's one of the best developed countries in `lsms_library`. But I encourage you to play with others; just change the string 'Uganda' below.

```
eg = ll.Country('Uganda')
eg.waves
```

What different kinds of features/data are available for this country?

```
eg.data_scheme
```

What does the sample look like?

```
eg.sample()
```

Some actual data:

```
eg.household_roster().sort_index()
```

1.7 Survey Design

Any household survey has:

Population What is the *population* that is supposed to be represented?

Sample A collection of selected households

Sample Frame What is the practical information we have on the population that we can use to construct the sample?

Sample Weights Proportional to the *ex ante* probability that a given sample household was selected for inclusion in the sample.

Sample Data Actual information collected from actual households in the sample

1.8 Survey Design Example

Consider the Ghana Living Standards Survey (GhanaLSS), and specifically the most recent publicly released wave 7 (2016–17).

Sample Frame A list of private dwellings from the 2010 census. What does this exclude?

Sample Weights Proportional to the inverse of the *ex ante* probability that a given sample household was selected for inclusion in the sample. The weights in `lsms_library` are normalized to have a mean of one, but sometimes instead they're normalized to sum to the estimated size of the population.

1.9 Survey Design Example

Consider the Ghana Living Standards Survey (GhanaLSS), and specifically the most recent publicly released wave 7 (2016–17).

Population

```
glss = ll.Country('GhanaLSS')
WAVE = '2016-17'

glss.population[WAVE].population_statement
```

Sample

```
glss7 = glss[WAVE]
glss7.sample()
```

Sample Frame A list of private dwellings from the 2010 census. What does this exclude?

Sample Weights Proportional to the inverse of the *ex ante* probability that a given sample household was selected for inclusion in the sample. The weights in `lsms_library` are normalized to have a mean of one, but sometimes instead they're normalized to sum to the estimated size of the population.

Sample Data Actual information collected from actual households in the sample. For example, here's some information on the individuals within each household in the GhanaLSS 2016-17 data:

```
glss7.household_roster()
```

There are many different similar tables available; to see them

```
glss.features
```

1.10 Choosing a Population

In designing a population survey, one has to first choose a population. We'll be focused on *household* surveys within a particular country.

That sounds like a pretty good start: for example, all households in Ghana. But almost immediately edge cases turn up.

- First, what's a household? Usual definition: a set of people who "share a cooking pot"
- What about people in Ghana who don't live in households?
  - Convicts who live in prisons
  - Students who live in dormitories
  - Nuns who live in monasteries
- **Typical solution**: "private households" (so we're excluding the above)

1.11 Choosing a Survey Frame

If we've settled on "private households", we have to figure out a way to actually somehow enumerate all the private households in a country so we can make a random selection.

- Possibilities
  - Use a census (the usual LSMS approach)
  - Use remote sensing and machine learning to try to identify private dwellings (Togo a recent example)
- Problems
  - Census is usually out of date
  - Remote sensing is highly error prone
- Leads to demand for ground-truthing

1.12 Census approach to enumerating households

Typical approach:

1. Divide entire country into "Enumeration Areas" (EAs).

2. Send a team to each EA; have them identify/map all residential structures (separate dwellings, apartment blocks, compounds).

3. Find someone who lives in each structure; if it's a private dwelling, list all who live there.

1.13 Census approach to enumerating households

Step 3 in full:

Find someone who lives in each structure; if it's a private dwelling, list all who live there. If it's an apartment block, get access, and list all who live in each apartment. If it's a compound with multiple dwellings: list regular occupants of each. Are cooking duties shared across dwellings? If so it's a single household; otherwise multiple HHs.

1.14 Two-Stage Cluster Based Sampling

At the time a census is conducted, it's usually the most reliable "ground-truth"; it provides a list of private households, you could just randomly select. But:

- Census out of date almost immediately; new construction, people move, etc.

1.14.1 Stages

1. Take a random sample of EAs. Then, *just for these sampled areas*, send a team to update the census listing, and get an up-to-date list of all households in just the *selected* EAs.

2. For each sampled EA, randomly sample the households within it.

1.15 Example: Two-Stage Sampling Weights

Guinea-Bissau provides a fairly simple example. It stratifies by region (see below); within a region it uses a simple two-stage sample. Consider the region Bafata; here is a map of the sample EAs chosen in the first stage.

```
ll.coordinate_map('Guinea-Bissau', where={'Region': 'bafata'})
```

1.16 Sampling Weights

We can always compute statistics (mean, variance, etc.) that are valid for the *sample*; but usually we'd like to use the sample to say something about the *population*. For this we need to know the probability π_i that a particular sample household i was included in the sample *ex ante*.

Once we know these we can compute statistics from a sample of size N.

For the sample:

Mean $\bar{x} = \frac{1}{N}\sum_{i\in\text{Sample}} x_i$

Variance $\text{Var}(x) = \frac{1}{N}\sum_{i\in\text{Sample}} x_i^2 - \bar{x}^2$

1.17 Sampling Weights

For the population

Let $\hat{\bar{X}} = \frac{\sum_{i\in\text{Sample}} x_i/\pi_i}{\sum_{i\in\text{Sample}} 1/\pi_i}$ (this is the Hájek estimator).

Mean $\bar{X} = \mathbb{E}\left[\hat{\bar{X}}\right] + O(1/N)$

Variance $\mathbb{E}(X-\bar{X})^2 = S^2 = \mathbb{E}\left[\frac{\sum_{i\in\text{Sample}}(x_i-\hat{\bar{X}})^2/\pi_i}{\sum_{i\in\text{Sample}} 1/\pi_i}\right] + O(1/N)$

Takeaway: To draw inferences about the population we need to know sampling probabilities π_i. *It is not enough to just have the data!*

1.18 Sub-Populations

Conducting a large-scale household survey is expensive! Sometimes you'd like to draw inferences not just about the original population (e.g., all the private households in Ghana), but also for subpopulations: e.g.,

- Particular regions
- Rural vs. Urban

The GLSS: Later rounds of the GhanaLSS are designed to do *both*; for example, to make population statements about the urban population of the Ashanti region.

1.19 Stratification

The most statistically efficient way to do inference for sub-populations is via **stratification**: Partition the overall population into different parts, and (effectively) treat each one as it's own sampling problem.

- Let N_s be the size of the sample for stratum s, and n_s be the size of the population in the stratum.

1.20 Strata Weights

If $N_s/n_s \neq N/n$ (the sample drawn in stratum s isn't proportional to the share of s in the population) then we've violated the maxim that every household in the overall population has an equal chance of being selected. To correct this, construct strata weights

$$v_s = \frac{N_s/n_s}{N/n}$$

to correct for the fact that a household in stratum s has a probability of being selected into the sample which must be adjusted by the factor v_s.

1.21 Two-Stage Sampling Weights

In a Two-Stage Clustered sampling scheme, the probability of a particular household i who lives in EA c_i being included in the sample depends on two other probabilities:

- The probability of EA (cluster) c_i being selected in the first stage, say p_{c_i};
- *Conditional* on c_i being selected, the probability of i being selected from all the other households in the cluster, say r_i.

So, in a two stage clustering scheme:

$$\pi_i = p_{c_i} r_i$$

1.22 Example: Two-Stage Sampling Weights

Back to our Bafata example.

```
ll.coordinate_map('Guinea-Bissau', size='weight', where={'Region': 'bafata'
```

```python
s = ll.Country("Guinea-Bissau").sample().reset_index()
b = s[s["strata"] == "Bafata"]

cl = b.groupby("v").agg(w=("weight", "first"),
                        urban=("Rural", "first"),
                        take=("weight", "size"))
cl["w"] = cl["w"].astype(float)

print("clusters  :", len(cl))
print("households:", len(b))
print("take      :", dict(cl["take"].value_counts()))
```

Fifty clusters, 598 households, and forty-eight of the fifty contain exactly twelve households — the two that hold eleven are non-response, not design. This is the textbook object: draw clusters, take a fixed number from each.

```python
cv = b.groupby("v")["weight"].std().fillna(0) / b.groupby("v")["weight"].me
print("clusters whose weight varies inside them:", int((cv > 1e-9).sum()),
print("weights: min %.3f  median %.3f  max %.3f" % (cl.w.min(), cl.w.median
```

The weight is constant within every cluster, but not *across* clusters.

- Everyone in cluster selected with equal probability
- But weights differ *across* clusters, which tells us different cluster are selected with *different* probabilities.

1.23 Inferring sample design from weights

For household i in cluster c_i, a two-stage design factors the probability of selection:

$$\pi_i = \underbrace{p_{c_i}}_{\text{cluster chosen}} \times \underbrace{\frac{b_{c_i}}{M_{c_i}}}_{r_i\text{: chosen within it}}$$

b_{c_i} is the number of households in the cluster selected (twelve); M_{c_i} is the size of the cluster. The weight is the reciprocal:

$$w_i \;\propto\; \frac{1}{\pi_i} \;=\; \frac{M_i}{p_i\, b_i}$$

One equation, two unknowns. We see w and b; we want p and M.

1.24 Inferring sample design from weights

```
print("sum of weights over the whole survey:", round(s["weight"].sum(), 1))
print("households in the survey            :", len(s))
```

Equal — the library normalizes weights to mean 1, so they carry *relative* probabilities and nothing about population totals. Everything below is a ratio.

Perhaps the fifty clusters were drawn by **simple random sampling** from Bafata's enumeration areas, each equally likely. Then $p_c = N/n$ is the same for all of them and the identity collapses:

$$w_i \propto \frac{M_i}{b_i}$$

A household in a large EA is less likely to be drawn, so it stands for more people.

```
cl["M_rel"] = cl["w"] * cl["take"]
cl["M_rel"] = cl["M_rel"] / cl["M_rel"].median()

print(cl["M_rel"].describe(percentiles=[.05, .25, .5, .75, .95]).round(2).t
```

Sanity check : Half the clusters fall between 0.89 and 1.04 times the median. Enumeration areas are usually *built* to be roughly equal in size.

1.25 Recovering both probabilities

```
cl["pi_rel"] = 1 / cl["w"]                    # overall, relative
cl["stage2"] = cl["take"] / cl["M_rel"]       # within-cluster, relative
cl["stage1"] = cl["pi_rel"] / cl["stage2"]    # what is left over

print("stage-1 probability across the 50 clusters:")
print("  cv = %.2e  (SRS says it should be constant)" % (cl["stage1"].std()
```

Zero to numerical precision. This is about the simplest case of two-stage cluster sampling:

1. EAs drawn with equal probability ($r_c = 1/\text{Number of EAs}$)
2. Households within a EA drawn with equal probability ($p_i = 1/M_{c_i}$)

1.26 Hazards of ignoring sample weights

Consider the following **Population Pyramid**, a nice graphical device for thinking about the demographic structure of a population. Because we're after population statistics, here we use the sample weights:

```
from lsms_library.visualizations import population_pyramid

ax = population_pyramid(glss, wave=WAVE, weights=True)
ax.figure.set_size_inches(7, 5.5)
```

1.27 Hazards of ignoring sample weights

Now, contrast if we *neglect* the survey weights, and just treat our data as if it was a simple random sample:

```
ax = population_pyramid(glss, wave=WAVE, weights=True, ghost=False)
ax.figure.set_size_inches(7, 5.5)
```

The filled bars are weighted; the outline is unweighted. The outline stands proud of the fill at every young age band and hugs it at the old ones, so the unweighted sample contains proportionally **more children** than the population it represents.

1.28 Why?

Let's consider two facts.

```
r = glss.household_roster(waves=[WAVE]).reset_index()
s = glss.sample(waves=[WAVE]).reset_index()

people = len(r)
weighted_total = s.set_index("i")["weight"].reindex(r["i"]).sum()
print(f"people in the roster      : {people:,}")
print(f"sum of their weights      : {weighted_total:,.1f}")
print(f"mean weight per household : {s['weight'].mean():.4f}")
```

The mean weight is 1 by construction. But that mean is over **households** while the roster is one row per **person**. So the weighted person-total is $\sum_h n_h w_h$, which equals the headcount only if weight and household size are uncorrelated.

So are they uncorrelated?

```
size = r.groupby("i").size().rename("hhsize")
d = s.set_index("i").join(size).dropna(subset=["hhsize"])
print("corr(weight, household size) =", round(d["weight"].corr(d["hhsize"])
print()
print(d.groupby(pd.cut(d.hhsize, [0, 1, 2, 3, 5, 8, 50],
                       labels=["1", "2", "3", "4-5", "6-8", "9+"]),
           observed=True)
      .agg(households=("weight", "size"), mean_weight=("weight", "mean"))
      .round(3).to_string())
```

Bigger households carry smaller weights. Since bigger households hold more children, an unweighted count over-represents the young.

2 Describing Welfare

2.1 Introduction

2.1.1 Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton, chs. 3 and 4
- Deaton & Zaidi, *Guidelines for Constructing Consumption Aggregates*
  - Deaton & Zaidi §§2–4 if you read nothing else
- **Christiaensen, Ligon, & Sohnesen (2022)**: at least memorize the title.

2.1.2 Measuring Individual Welfare

Many notions of individual welfare; generally lead to some unidimensional index, say w (bigger is better):

- Subjective ("How happy are you?")
- Utility (How is this different from Subjective happiness?)
- Material (Something about command over economic resources)
  - Wealth?
  - Income?
  - Consumption expenditures?

1. Aside: What about risk?

2.1.3 Material Well-Being: Statistics

Some summary statistic of the *distribution* F of w in the population.

Inequality E.g., Atkinson's (1971) utilitarian notion of inequality:

$$I = U\left(\int_{\underline{w}}^{\bar{w}} w dF(w)\right) - \int_{\underline{w}}^{\bar{w}} U(w) dF(w) \geq 0$$

Poverty A quantile of some function $\phi_z(w)$; e.g.,

$$P(z) = \int_{\underline{w}}^{z} \phi_z(w) dF(w)$$

These are distributions *of the population*, so in practice those sample weights will matter!

2.1.4 Conventional options for w

1. Income
   - Difficult to measure if sources are agricultural or informal
   - Also highly variable—a farmer's main income may arrive once a year, after harvest, and be very risky.
2. Wealth
   - What is it?
   - Perhaps something like discounted expected income?
   - But if income is difficult to measure then all possible future incomes may be much more difficult

2.1.5 Conventional options for w, continued

1. Assets
   - Related to wealth, but may be measurable now
   - Financial assets + Bicycles + Goats + House = ?? In principle these could be priced, but this may not be at all practical.
2. Consumption Expenditures The "Gold standard"
   - Probably easier to measure than income
   - Consumption is smoother than income
   - And closer to the theoretical object (permanent income, lifetime resources) that we really want.

2.2 The Theoretical Object

2.2.1 The household's problem

What is the consumption aggregate a measure *of*? Assume:

(A1) Preferences $U(c)$ over J goods, increasing, strictly concave, the same every period, discounted at β.

(A2) One decision maker the household maximizes a single U.

(A3) One asset a risk-free bond, gross return R, no borrowing limit.

(A4) Expectations $\mathbb{E}_t$ over the true conditional distribution.

With A_t resources on hand and $x_t = p_t \cdot c_t$ total expenditure,

$$V_t(A_t, p_t) = \max_{c_t} U(c_t) + \beta \mathbb{E}_t V_{t+1}(A_{t+1}, p_{t+1}),$$

subject to $A_{t+1} = R(A_t - x_t) + y_{t+1}$.

2.2.2 The marginal utility of expenditure

Differentiate. With $u_j = \partial U / \partial c_j$, the first-order condition in c_j and the envelope condition in A are

$$u_j(c_t) = \lambda_t \, p_{tj}, \qquad \lambda_t \equiv \beta R \, \mathbb{E}_t[\partial V_{t+1}/\partial A] = \partial V_t / \partial A.$$

So λ_t is two things at once, which is what makes it useful:

- the multiplier on the period budget constraint, the utility value of one more cedi spent *today*;
- the derivative of the *value function*, the utility value of one more cedi of resources, the whole future included.

V is concave in A, so λ falls as the household gets richer. Write $w_t = -\log \lambda_t$, which therefore rises with welfare. This is another candidate for the w of the opening slide.

2.2.3 The martingale condition

Substituting the envelope condition into the definition of λ_t,

$$\lambda_t = \beta R\, \mathbb{E}_t \lambda_{t+1}.$$

When $\beta R = 1$ the marginal utility of expenditure is a martingale:

$$\mathbb{E}_t \lambda_{t+1} = \lambda_t.$$

A statement about information: whatever the household knew at t is already in λ_t, so nothing dated t forecasts the change.

- Needs A1–A4 *and* an interior solution: no binding borrowing limit.
- With several assets, $\beta \mathbb{E}_t[(\lambda_{t+1}/\lambda_t) R^j_{t+1}] = 1$ for every asset j held in positive quantity.

2.2.4 The martingale condition

Three things this doesn't say, since all three get attributed to it.

It isn't a statement about consumption. It's a statement about marginal utility, and getting from the one to the other takes a further assumption, which is the business of the next two slides.

It isn't a statement about $\log \lambda$. Jensen's inequality runs the wrong way: if λ is a martingale then $\mathbb{E}_t \log \lambda_{t+1} < \log \lambda_t$, so the welfare index $w = -\log \lambda$ drifts upward even when the model holds exactly. Anyone who regresses Δw on lagged information and recovers a constant has found the constant, not a violation.

And it isn't a statement about levels. $\beta R = 1$ is a knife edge adopted for convenience. Impatient households have $\beta R < 1$, so $\mathbb{E}_t \lambda_{t+1} = \lambda_t / (\beta R)$ exceeds λ_t: marginal utility drifts up, and consumption falls across the life-cycle.

Where the condition fails is more interesting than where it holds. Suppose the household would like to borrow and can't. Then A3 is false, the first-order condition holds as an inequality, and $\lambda_t > \beta R\, \mathbb{E}_t \lambda_{t+1}$: current marginal utility is too high relative to future, which is what being constrained means. Distinguishing that household from one that is simply saving against future risk is hard, since both look like a household that isn't spending. One approach elicits the intertemporal price $q = \mathbb{E}_t[\beta \lambda_{t+1}/\lambda_t]$ directly and asks how it moves in response to a windfall: up if a constraint was binding, down if precaution dominates. Not something an LSMS will tell you.

2.2.5 Why total expenditure?

Three steps, each with its own assumption. Let $v(x,p) = \max\{U(c) : p \cdot c \leq x\}$, so $\lambda = \partial v / \partial x$.

1. At *given* prices x orders welfare, since $\lambda > 0$. Nothing beyond A1 is needed, but this ranks within a market only.

2. To rank across markets we need a scalar $P(p)$ with $v(x,p) = \psi(x/P(p))$. Such a P exists if and only if U is homothetic (A5), and then $P = e(p)$, the cost of a unit of the aggregate. Quasi-homothetic (A5$'$): two indices, a line $a(p)$ and a deflator $b(p)$.

3. So $\lambda = \psi'(x/e(p))/e(p)$, decreasing in real expenditure.

Under A5, real total expenditure $x/e(p)$ is a sufficient statistic for within-period welfare. It is what the rest of this session builds.

2.2.6 Why total expenditure?

A5 is doing more work than it appears to. It is exactly the license to divide nominal expenditure by a price index, which is what the deflation slide does and what every poverty statistic in this session does after it. Better to carry it as a stated assumption than as a technicality.

It can be weakened, and the weaker version is the one this session's practice already presumes. Suppose preferences are *quasi*-homothetic (A5$'$): the expenditure function is Gorman's $e(u,p) = a(p) + b(p)u$, so $v(x,p) = \psi\big((x - a(p))/b(p)\big)$ and demands are linear in x. Two price indices now do the work of one, and both have names this session already uses. $a(p)$ is the cost of a fixed bundle at local prices, which is what a poverty line is; $b(p)$ deflates whatever is spent above it. A5 is the case $a = 0$. Under the iso-elastic cardinalization of the next paragraph,

$$w = \frac{1}{\sigma} \log \frac{x - a(p)}{b(p)} + \log b(p),$$

log expenditure above the line. So an absolute line plus a deflator, the procedure the rest of this session teaches, is exact for a quasi-homothetic system rather than an approximation to a homothetic one; and unlike A5 it is consistent with Engel's law, since Gorman Engel curves are linear in x without being proportional to it. What it is not consistent with is a demand system of rank three, which is where session 3 goes.

Getting from an ordering to a number takes one more choice. Ordinal preferences fix v only up to an increasing transform, and λ is not invariant to that transform, so a cardinalization has to be picked. Take the iso-elastic one, (A6): $\psi(G) = G^{1-1/\sigma}/(1-1/\sigma)$. Then $\lambda = (x/e(p))^{-1/\sigma}/e(p)$, so that

$$w = -\log\lambda = \frac{1}{\sigma}\log\frac{x}{e(p)} + \log e(p).$$

Within a market $e(p)$ is common to everyone, so w is an increasing affine function of log real expenditure, and two households in the same cluster are ranked identically by the marginal utility of expenditure and by log consumption. Across markets the price term enters additively. That equivalence is worth holding onto: the elaborate object and the simple one agree, *given* A5 and A6.

Durables take a separate assumption, (A7). A refrigerator is an asset, not a meal, so buying one is a portfolio transaction, and the single bond of A3 hides it. What belongs in U is the flow of services, which is why the aggregate imputes a use value rather than recording the purchase. Handling the rest of the bundle without tracking the durable stock alongside it requires the services of durables to be weakly separable from nondurable consumption. Assumed, not tested, and it is the reason the aggregate is described as covering nondurables and services.

2.2.7 Hall's random walk

The strict permanent income hypothesis adds:

- **(A8) Constant prices** or a martingale in p independent of λ, which puts the walk in quantities.
- **(A9) Quadratic utility** $U(x) = -\frac{1}{2}(\bar{x}-x)^2$, so $\lambda_t = \bar{x} - x_t$ is *linear* in expenditure.
- **(A10)** $\beta R = 1$ with R constant and known.

The martingale in λ is then a martingale in x:

$$\mathbb{E}_t[\bar{x} - x_{t+1}] = \bar{x} - x_t \quad \Longrightarrow \quad \mathbb{E}_t x_{t+1} = x_t.$$

Consumption is a random walk (Hall 1978): nothing dated t forecasts Δx_{t+1}, lagged income included. Testable, but it needs longitudinal data.

2.2.8 Hall's random walk

A9 should bother you, though not for the reason usually given. Quadratic utility isn't homothetic, but it is *quasi*-homothetic: demands are linear in x, the expenditure function is Gorman's $a(p)+b(p)\tilde{u}$, and that is the weakened A5 of the previous aside, so the two sit together perfectly well. What should bother you is which member of the class it is. Its $a(p)=p\cdot\bar{c}$ is the cost of the *bliss* bundle, with expenditure below it, where the Stone–Geary members put $a(p)$ at the cost of a subsistence bundle and expenditure above it; it has increasing absolute risk aversion, so richer households take less risk in absolute terms, which is backwards; and marginal utility reaches zero at $\bar{c}$. It is adopted for exactly one property, that marginal utility is linear in x, which is what turns a martingale in λ into a martingale in x. Certainty equivalence is a feature of the quadratic, not of the world.

The one-good form on the slide is for legibility, not necessity. Take a vector c and any positive definite A, and $U(c)=-\frac{1}{2}(\bar{c}-c)'A(\bar{c}-c)$. The first-order condition $A(\bar{c}-c)=\lambda p$ and the budget give

$$\lambda=\frac{a(p)-x}{b(p)^2},\qquad a(p)=p\cdot\bar{c},\quad b(p)^2=p'A^{-1}p,$$

which is linear in x at *any* prices, with the same a and b as the Gorman form above. So A8 is not a claim about how many goods there are. Nor, strictly, is it a claim that prices are constant: that is sufficient, since it makes a and b constants, but what the argument actually uses is that $\lambda_t p_t$ be a martingale, and prices that are themselves a martingale, with innovations independent of those in λ, deliver that too. The demands are $c_t=\bar{c}-\lambda_t A^{-1}p_t$, linear in λp, so the *quantity* vector is then a martingale, $\mathbb{E}_t c_{t+1}=c_t$, and so is consumption valued at any fixed base prices; which is Hall's object, since he had one good. What is not a martingale, even then, is nominal expenditure. At given λ, $x=p\cdot\bar{c}-\lambda\, p'A^{-1}p$ is concave in p, so a mean-preserving spread in prices lowers expected spending: $\mathbb{E}_t x_{t+1}=x_t-\lambda_t\,\mathrm{tr}(A^{-1}\Sigma_p)$. Second order in price volatility, but a drift, and downward. A panel test of the random walk should therefore be run on consumption at constant prices rather than on nominal expenditure, which is the usual practice anyway and now has a reason.

Swap A9 for the iso-elastic A6 and the conclusion changes. The martingale is in $\lambda=(x/e)^{-1/\sigma}$, so $\mathbb{E}_t x_{t+1}^{-1/\sigma}=x_t^{-1/\sigma}$, and Jensen now delivers $\mathbb{E}_t x_{t+1}>x_t$ whenever next period's expenditure is risky. Consumption drifts up, because the household accumulates a buffer and runs it down slowly. That is precautionary saving, and it lives in the third derivative of U, which the quadratic sets to zero. The random walk didn't survive a change of cardinalization, which is a reason to ask what else doesn't.

The empirical record is worse than that. Consumption growth is predictable from lagged income in most panels that have looked, the "excess sensitivity" of Zeldes (1989) and a large literature since, and the buffer-stock models of Deaton (1991) and Carroll (1997) fit low-income data considerably better than the strict version does.

What survives is the weak claim, and it is enough. The marginal utility of expenditures/assets λ_t is forward looking because it is the derivative of a value function, and it stays forward looking whether or not $\beta R = 1$ and whether or not utility is quadratic. A number built from this period's consumption therefore carries the household's own view of its prospects in a way that a number built from this period's income does not. That is the entire case for what follows. It is a good deal weaker than "consumption is a random walk," and fortunately it is all we need.

2.2.9 The assumptions, collected

	Assumption	What it buys
A1–A2	Time-separable U, one maximizer	a well-defined λ
A3	One asset, no borrowing limit	an interior solution
A4	Rational expectations	$\mathbb{E}_t$ is the truth
A5	Homotheticity, or quasi-	a deflator, or a line and one
A6	Iso-elastic cardinalization	w affine in log real x
A7	Durables weakly separable	nondurables and services
A8–A10	Constant p, quadratic U, $\beta R = 1$	Hall's random walk

Everything from here to the end of the session is an attempt to measure x. None of it repairs an assumption in this table.

2.3 Practical Difficulties in Measuring Individual Items of Consumption

2.3.1 The four components of consumption expenditures

Or: "Nondurable consumption and services".

Consumption expenditures in the LSMS surveys generally fall into one of five categories, and are typically elicited in different modules.

1. **Food** — purchased, own-produced, and received as gifts

2. **Non-food, non-durable** — fuel, transport, clothing, school fees, health

3. **Durables** — *value of services*, or "use value", not purchase price

4. **Housing** — rent, actual or imputed

5. **Other Services** — Not really measured at all

2.3.2 List Length

Beegle et al (2012): the *number* of goods asked has a large effect on recall, and on that evidence many NSOs have lengthened the list. Just for food:

Ethiopia	25 → 26 → 55 → 73 → 74
GhanaLSS	58 → 58 → 117 → 112 → 114 → 114 → 179
Burkina Faso	55 → 130 → 152
Uganda	61 → 63 → 70 → 71 → 77 → 77 → 117 → 116
Nigeria	90 → 90 → 96 → 104 → 116 → 116 → 110 → 110

- Should make measured total consumption more accurate;
- BUT: Really complicates comparisons across time.
- And: Can greatly increase cost of fielding the survey.

2.3.3 Timing & Recall

The model treats time as discrete. What in the field corresponds to a "period"? LSMS practice reveals what the designers thought.

Food Usually a seven-day recall, with interesting variation and some explicit experiments (Beegle et al. 2012) on how well respondents remember purchases.

Non-food expenditures or "semi-durables" Think things like clothing, inventories of kerosene.... Often a 30-day recall.

Durables Bicycles, radios, fans,.... Often a year-long recall for purchases, and/or an inventory of the *stock* of durables.

2.3.4 Food prices: which price?

Food *expenditures* are quantities times prices. Which prices?

- farmgate price received — what the household gave up
- local market price — what it would have cost
- these differ by the marketing margin, said to be 30–50%

Deaton & Zaidi: value at the price the household *faces*. In practice, a median unit value by item, cluster and round, falling back up the hierarchy when cells are thin.

Which of the two does a survey's own valuation question measure?

2.3.5 Food prices: what GLSS7 asks, and what it gets

Section 8 Part H, question 9: **"for how much would you sell one unit of ... now?"** A farmgate question, by its wording.

Against what neighbours *paid*, same item, unit and cluster:

- Ghana 2016-17, 322 cells: median ratio **1.000**; a third tie exactly
- Uganda, eight waves, the same question: **1.000** pooled
- Uganda also records sales: the valuation is **1.23** times the sale price; buyers paid **1.60** times it

Asked what a bowl of maize would sell for, a household answers with the price of a bowl of maize. The margin is real — 1.3 to 2.2 where a survey records sales — and this question does not measure it.

2.3.6 Food prices: which price?

Start with what is actually there. Four sources get named in discussions like this one — unit price, market price, farmgate price, community price — and GhanaLSS 2016-17 has three, of which two of the names refer to the same instrument. `food_prices()` is the unit value implied by purchases, and it is flagged `purchased` and nothing else. `community_prices()` is the community questionnaire, priced at the cluster, which is both the "market" and the "community" price of the list. And own-production rows in `food_acquired()` carry a price the household itself gave, elicited as question 9 of Section 8 Part H: "for how much would you sell one unit of ... now?" By its wording that is a farmgate question, what the household would *receive*, so the comparison the slide sets up ought to be available.

Measured, it is not a farmgate price. Put the household's valuation beside the unit value its neighbours paid for the same item, in the same unit, in the same cluster, keeping only cells with at least ten observations on each side, and the median ratio across 322 cells is 1.000; 31.7% of cells agree to the pesewa. Nor is this a Ghanaian peculiarity. Uganda asks the same question in eight waves, and pooling the two countries gives 1,666 cells with a median ratio of 1.000. Asked what a bowl of maize would sell for, a household answers with the price of a bowl of maize.

Uganda is where to see what that costs, because it is the one survey that also records actual crop sales. On the 223 cells where all three prices exist, buyers paid 1.60 times what sales fetched — there is the marketing margin — and the household's valuation was 1.23 times the sale price, in between and no nearer one end than the other. The interviewer manual there says in so many words that the valuation is to be at the farm-gate price, excluding transport and marketing. It is not. So an aggregate that values own consumption at the price in `food_acquired` is valuing it about 23% above what the household would actually have received, and carries part of a margin the questionnaire meant to exclude. Ghana records no sales, so the same number cannot be produced here; but the valuation behaves the same way, and there is no reason to expect the consequence to differ.

The margin itself is real. Where a survey records both a sale and a purchase of the same good in the same container, the ratio runs from 1.30 in Ethiopia through 1.50 in Malawi and 1.52 in Nigeria to 1.57 in Uganda and 2.24 in Tanzania: the textbook's 30–50% and then some. What the textbook does not say is that the question surveys use to elicit a producer price does not measure one.

The community questionnaire tells a weaker version of the same story. Its price sits 20% above the household's valuation in 2016-17, 5% and 109% above it in 1991-92 and 1998-99, and 13% below it in 2012-13, on a few dozen cells each, so there is no stable margin there either; the enumerator weighs what a vendor sells and the household reports a container, and the two units seldom mean the same thing. Whatever the household reports, it tracks what its neighbours pay far more closely than what a vendor charges.

What follows for practice. Deaton and Zaidi treat the choice between the producer price and the market price as the analyst's. It is not: the respondents have made it, and they chose the market. Value own production at `food_acquired.Price` if you like, since at the median it is the price the household faces and that is what the guidelines ask for; but do not call it a farmgate price, and do not expect it to have moved your aggregate down by a third. If you need what production would fetch, you need a survey that records sales, and GLSS7 is not one.

The numbers above are from a study of eight countries' price sources in `LSMS_Library`, with the cells built to a common protocol; `price_sources.ipynb` builds Ghana's three prices at a simpler grain and asks what the data can and cannot say about them.

2.3.7 Food quantities, and what was never bought

Much of the food in an LSMS household was never bought:

- Purchased at market
- Consumed out of own production
- Gift/Transfer
- Consumed outside the household (?)

Not a canard. In Ghana 2016-17 own production is **28.7%** of the value of food; 12,191 of 13,942 households record some. Deaton and Zaidi's Table 3.1 implies 31.9%.

But it reaches your aggregate only if you *value* it: from 1991-92 on the survey records own production in quantities, not cedis.

2.3.8 Food quantities, and what was never bought

The claim survives contact with the data, which is not something one can assume of a claim repeated as often as this one. Valuing own production at the price the household itself reported, the share of food value that was never bought runs across the seven Ghanaian waves as

wave	1987-88	1988-89	1991-92	1998-99	2005-06	2012-13	2016-17
%	17.4	18.1	30.4	30.5	14.9	27.6	28.7

with no secular decline in sight. The 2005-06 figure is the one to be suspicious of, and chasing it down would make a decent exercise on its own.

Two things are worth knowing before you trust the 28.7%. The first is that the price is the household's own, not an imputation of ours, and it varies household by household within an item and unit: seven distinct values in the median item-unit cell, seventy-seven in the worst. So it is real reported data. Comparing it against the unit values implied by *purchases* of the same item in the same unit, the median ratio across 175 cells is 1.00, which is what the previous aside leads you to expect: the household is quoting the market price. The tails are not reassuring at all, running from 0.5 at the tenth percentile to 3.8 at the ninetieth, so the aggregate is sound and any individual household's imputation may not be.

The second is that 28.7% is a floor. In 2016-17 the survey records a price on 151,677 own-production rows but a quantity on only 138,787 of them, and a row missing either one contributes nothing to the product. Note also that GhanaLSS distinguishes two sources, purchased and produced, so the gifts and the meals eaten away from home in the list above are nowhere in this particular table. Both omissions push the same way.

The form that produces these numbers is in your `reading/` folder as `GLSS7-2016-17-Section8H-own-produce.pdf`: quantity consumed at each of the six visits, and one price, asked as what a unit would sell for now. The exercise notebook `food_sources.ipynb` takes this apart, and asks whether the pattern holds in the countries you care about.

2.3.9 Building the aggregate

`LSMS_Library` does the food side for you. Start from what was acquired:

```
# Show tracebacks without the library's internal frames: the line that
# failed, and why.  Change Plain to Verbose if you ever want the rest.
%xmode Plain

# The first call that builds a table may print "DVC unavailable ...
# falling back to manual aggregation".  That is about how the library
# fetches its raw files, not about your data; the numbers are the same,
# and it does not recur once the table is built.

# These notebooks are read two ways: on your own screen, and projected onto
# a wall in a lit room where thin lines and pale markers are simply not
# there.  Those want different plots, so the difference is a switch rather
# than a compromise that suits neither.  Leave it False; the lecturer flips
# it.  Everything below the 'if' is about the wall, not about your laptop.
PROJECTOR = False

import matplotlib as mpl
mpl.rcParams.update({'figure.dpi': 110, 'font.size': 11,
                     'lines.linewidth': 1.8})
```

```
if PROJECTOR:
    mpl.rcParams.update({
        'figure.figsize': (7, 4.5),
        'font.size': 15, 'axes.labelsize': 15, 'axes.titlesize': 17,
        'xtick.labelsize': 13, 'ytick.labelsize': 13, 'legend.fontsize': 13
        'lines.linewidth': 3.0, 'lines.markersize': 9, 'axes.linewidth': 1.
    })

import lsms_library as ll
import numpy as np, pandas as pd

ghana = ll.Country('GhanaLSS')
acq = ghana.food_acquired()
acq.head()
```

The index is $(t, i, j, u, s, visit)$ — wave, household, item, unit, source, visit — and the columns are `Expenditure`, `Quantity`, `Price`. The derived table collapses this to expenditure per household-item:

```
food = ghana.food_expenditures()
food.head()
```

Summing over items gives food *purchases* per household. Not consumption:

```
purchases = food.groupby(['t', 'i']).sum().squeeze()
purchases.groupby('t').describe().round(0)
```

`food_expenditures()` returns what households *bought*. Food they grew and ate never had an expenditure attached to it, so it is missing from that table — missing, not zero. The acquisition table records it differently:

```
acq.groupby(['t', 's'])[['Quantity', 'Expenditure', 'Price']].count()
```

Read that table carefully, because it is the whole problem in one picture. From 1991-92 on, own production carries a `Quantity` and a `Price` but no `Expenditure`, while purchases carry a `Quantity` and an `Expenditure` but no `Price`. That asymmetry is in the instrument. Section 8H (PDF, opens in a new tab) asks the quantity of own produce consumed at each of six visits and, in question 9, one price; the purchase form, Section 9B, further down, asks amount and quantity and never a price.

GLSS7 — Section 8 Part H: Consumption of own produce

Ghana Living Standards Survey round 7 (2016/17), household questionnaire Module B: how food the household grew and ate is elicited. Questions 3 to 8 ask the QUANTITY consumed since the last visit, once per visit, and never an amount spent. Question 9 asks instead "for how much would you sell one unit of … now?" — a price the household would receive, not one it paid. Source: Ghana Statistical Service, *GLSS7_MODULE B _14_10_16.docx*, from www2.statsghana.gov.gh catalog 97. Transcribed from the Word original for teaching; this is not a scan of the printed form.

NAME OF PERSON RESPONSIBLE ID CODE

RESPONDENT (COPY FROM SECTION 6 Q.7) | PERSON INTERVIEWED ID

2nd 3rd 4th 5th 6th 7th

day month day month day month day month day month day month

ITEM	CODE	1 Did the household consume any own produced ……….. in the past 12 months? Yes… .1 No……2 Next Item	2 How many months altogether was own produced ……….. consumed during the past 12 months?	3 How much of own produced ……. was consumed by the household since my last visit? (>> 9)		4 How much of own produced …….. was consumed by the household since my last visit?		5 How much of own produced …….. was consumed by the household since my last visit?		6 How much of own produced ……. was consumed by the household since my last visit?		7 How much of own produced ……. was consumed by the household since my last visit?		8 How much of own produced …….. was consumed by the household since my last visit?		9 For how much would you sell one unit of ………….now? **USE MOST OCCURING UNIT OF MEASURE**		**UNIT CODES REFER TO PAGE 8.10 OR REFER TO CODE BOOK**
			NO. OF MONTHS	QUAN-TITY	UNIT CODE	QUAN-TITY	UNIT CODE	QUAN-TITY	UNIT CODE	QUAN-TITY	UNIT CODE	QUAN-TITY	UNIT CODE	QUAN-TITY	UNIT CODE	AMOUNT GH¢	GHp	
GRAINS AND FLOURS																		
Rice(paddy, grain)	001																	
Maize-cob (fresh)	002																	
Maize-flour/dough	003																	
Sorghum/ guinea corn	004																	
Millet grain	005																	
Millet flour	006																	
Other grains	008																	
Other flours	009																	

The valuation Deaton and Zaidi spend a chapter on is, in this survey, a multiplication nobody performed:

```
value = acq.Expenditure.where(acq.Expenditure.notna(), acq.Quantity * acq.P
by_source = value.groupby(['t', 's']).sum().unstack('s')
(100 * by_source['produced'] / by_source.sum(axis=1)).round(1)
```

Own production is 28.7% of the value of food in 2016-17. So perform the multiplication. `value` already has it, expenditure where the household paid and quantity times its own price where it did not; summed over items it is food consumption per household, purchased and own-produced, and it is the aggregate the rest of this session uses. The library will do the same for you with `food_expenditures(basis`'total')=; here it is done by hand so that you can see exactly what was added.

```
food_total = value.groupby(['t', 'i']).sum()
pd.DataFrame({'purchases': purchases.groupby('t').median(),
              'purchased + own-produced': food_total.groupby('t').median()}
```

The median household's food consumption is well above its food purchases in every wave, and the gap is what the obvious call would have thrown away.

2.3.10 GLSS7's answer: a diary, visited six times

The most recent GhanaLSS is ambitious about frequent purchases: a diary, and an enumerator visiting six times at five-day intervals.

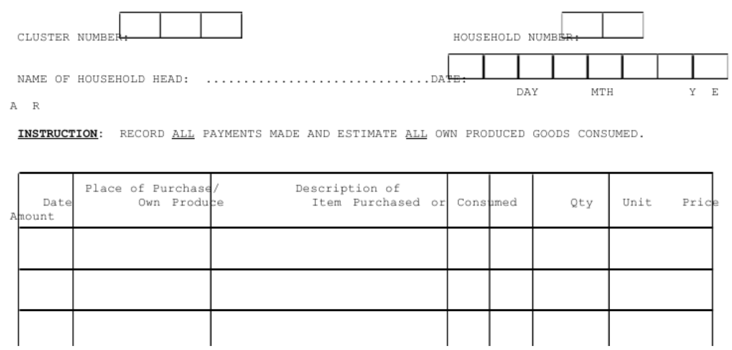
The form (`reading/`): six visit columns headed *2nd* to *7th*, asking **"...since my last visit?"**, and no price column anywhere.

Expensive; two experiments have asked whether it is worth it.

Scott & Amenuvegbe (1990) Ghana, and the reason GLSS asks this way

Beegle et al. (2012) Tanzania, eight modules, randomly assigned

2.3.11 The diary



CLUSTER NUMBER: HOUSEHOLD NUMBER:

NAME OF HOUSEHOLD HEAD: .... DATE: DAY MTH YEAR

INSTRUCTION: RECORD ALL PAYMENTS MADE AND ESTIMATE ALL OWN PRODUCED GOODS CONSUMED.

Date	Place of Purchase/ Own Produce	Description of Item Purchased or Consumed		Qty	Unit	Price Amount

"Record all payments made and *estimate all own produced goods consumed*": the diary asks for a price on own produce. Section 9B, filled from it, does not.

2.3.12 GLSS7's answer: a diary, visited six times

Worth putting the form beside the data, because it settles two things the code can only suggest.

The first is why purchased food has no price. The column headings are AMT, QTY and UNIT, repeated for each of the six visits, and there is nowhere to write a price. So the unit value you compute later is an arithmetic consequence of the form, not a choice anyone made, and the asymmetry with own production, which is asked at a price and never at an expenditure, is built into two different instruments rather than into the processing.

The second is the wording, "since my last visit". Every visit asks it, so the recall window is whatever the enumerator's schedule made it, and the household is not told what period to think about. Five days is the design; the form does not say five days. That is also the most likely explanation for the first visit sitting 20% above the rest, since the gap before the first expenditure visit is the one that is neither five days nor known. The exercise notebook asks you to test that against a longer-window story, and the form is why the question is worth asking rather than being a fishing expedition.

It is the same phrase, incidentally, that the 1987-88 and 1988-89 rounds record as their entire recall period. The wording did not change; what changed is that later rounds pinned it down with a visit schedule.

2.3.13 GLSS7's six visits

This is the form the diary is kept on, Section 9B: six visit columns headed *2nd* to *7th*, AMT, QTY and UNIT under each, and nowhere to write a price. Open it beside the code: Section 9B (PDF, opens in a new tab; also `reading/` in your home directory), and the booklet the household keeps between visits, which 9B is filled from: the diary.

GLSS7 — Section 9 Part B: Expenses (frequently purchased items)

Ghana Living Standards Survey round 7 (2016/17), household questionnaire Module B: the opening of the food expenditure elicitation. Note the six visit columns (2nd–7th), the recall wording "since my last visit", and that the answer columns are AMT, QTY and UNIT: there is nowhere to record a price. Source: Ghana Statistical Service, *GLSS7_MODULE B _14_10_16.docx*, from www2.statsghana.gov.gh catalog 97. Transcribed from the Word original for teaching; this is not a scan of the printed form.

	ID
PERSON RESPONSIBLE	
COPY FROM SECTION 6 Q.8	
2nd 3rd 4th 5th 6th 7th	

VISIT

day month day month day month day month day month day month

ITEM	CODE	1. How much was spent and what quantity was acquired of...........since my last visit?			2. How much was spent and what quantity was acquired of...........since my last visit?			3. How much was spent and what quantity was acquired of...........since my last visit?			4. How much was spent and what quantity was acquired of...........since my last visit?			5. How much was spent and what quantity was acquired of...........since my last visit?			6. How much was spent and what quantity was acquired of...........since my last visit?		
		AMT (GH¢)	QTY	UNIT	AMT (GH¢)	QTY	UNIT	AMT (GH¢)	QTY	UNIT	AMT (GH¢)	QTY	UNIT	AMT (GH¢)	QTY	UNIT	AMT (GH¢)	QTY	UNIT
FOOD AND NON-ALCOHOLIC BEVERAGES																			
Rice - Local																			
Dada Ba	001																		
Oman Ba	002																		
Mandela	003																		
Gold Star	004																		
Afife	005																		
Other local(specify)	006																		
Rice - Imported																			
Jasmine Rice	010																		
Basmati Rice	011																		
Uncle Ben's Rice (imported)	012																		
Lele Rice	013																		
Royal Feast	014																		
Gino Rice	015																		
Royal Aroma	016																		
Heaven	017																		
Cindy	018																		
Bella	019																		
Sultana	020																		
Mama's Pride	021																		
Red Eagle	022																		
Other imported rice specify	023																		
Guinea corn/sorghum	027																		
Maize																			
White maize grains	028																		
Yellow maize grains	029																		
Millet	030																		
Millet flour	031																		

The diary design is visible in the data. GLSS7 carries a `visit` level:

```
glss7 = ghana.food_acquired().xs('2016-17', level='t').xs('purchased', leve
sorted(glss7.index.get_level_values('visit').unique())
```

Six of them, five days apart. The form numbers them 2 to 7, because the food questions begin at the second visit; the library on the hub labels them 1 to 6, and other versions use the form's numbers. The text below uses the form's numbering, and the code selects the first food visit by position so that it does not care. Each visit covers the same five days, so if the instrument worked perfectly they should all look alike. They do not:

```
byvisit = pd.DataFrame({
    'exp':   glss7.groupby(['i', 'visit'])['Expenditure'].sum(),
    'items': glss7.groupby(['i', 'visit']).size(),
})
(byvisit.groupby('visit')
        .agg(mean_exp=('exp', 'mean'), mean_items=('items', 'mean'))
        .assign(exp_per_item=lambda d: d.mean_exp / d.mean_items)
        .round(2))
```

The first visit stands well above the five that follow, and then the series is flat. That is the shape Scott and Amenuvegbe found: steep early, flat later. Which of two stories is it, though? More items remembered, or bigger amounts attached to the same items?

```
v1 = sorted(byvisit.index.get_level_values('visit').unique())[0]
# first food visit
first, rest = byvisit.xs(v1, level='visit'), byvisit.drop(v1, level='visit'
for col in ('exp', 'items'):
    print(f"{col:6s} first food visit vs the other five: "
          f"{100 * (first[col].mean() / rest[col].mean() - 1):+.1f}%")
```

Expenditure is 20.6% higher; the item count only 4.5%. So it is mostly larger amounts per item rather than fuller enumeration, which is what telescoping looks like: purchases from before the window get pulled into it. It is the start-up bias Scott and Amenuvegbe had to strip out before they could measure decay at all, and here it is in the survey you are about to build an aggregate from.

Whether the first visit is worth keeping is a real question, and `recall_and_diaries.ipynb` is where you get to argue about it.

2.3.14 How fast does recall decay?

Scott & Amenuvegbe (1990), 13 frequently purchased items in Ghana:

- Reported daily expenditure falls steadily as the recall window lengthens from one day to one week, then flattens between one and two weeks

- Roughly **2.9% lost for every day added**, after reweighting away purchase cycles and start-up bias
- The decline is larger for items bought more often
- A one-year recall is simply erratic

So a weekly recall understates frequent purchases by around **20%** — and that was, in their words, "an abnormally well controlled field situation."

2.3.15 How fast does recall decay?

The shape matters as much as the size. The decay is steep early and flat later, not linear, which is why the choice between a one-day and a one-week window costs far more than the choice between one week and two. A survey designer reading only the headline number would conclude that shorter is always better and keep shortening; the curve says the returns are nearly exhausted after a week, and that the remaining error is not recall at all.

It is worth dwelling on that last phrase, "an abnormally well controlled field situation." Scott and Amenuvegbe are describing their own experiment, not a production survey, and they raise the 20% as a lower bound on what a routine operation should expect. Anyone quoting the figure as *the* recall bias is quoting it a good deal more confidently than its authors did.

2.3.16 What actually drives the error

Beegle et al. (2012): eight modules randomly assigned across ~4,000 Tanzanian households, benchmarked against a personal diary.

- Recall *length* is the small knob: 7 vs 14 days moves food spending **7%**
- Item *list* design is the big one: **21%** between 58 items and 11 categories
- "Usual month" fails outright, and contaminates the module after it
- Household diaries miss about **19%** of what personal diaries catch

Same households: poverty from **47.5% to 66.8%**, by module alone.

2.3.17 What actually drives the error

The design deserves stating, because the comparisons only mean something against it. Five recall modules and three diary modules: a 14-day long list of 58 items; the same list at 7 days; a 7-day subset of 17 items covering about 77% of food expenditure and then scaled up; a 7-day list collapsed from 58 items into 11 categories; a "usual month" asked over 12 months; a household diary with frequent supervision; the same with infrequent supervision; and a personal diary, frequently supervised, as the benchmark.

Three findings are worth carrying away.

The first is that the profession has been tuning the wrong parameter. Recall length is what gets argued about, and it is worth about 7%; list design is what gets treated as an administrative detail, and it is worth three times as much. The subset list is the practical discovery here: 17 items, rescaled, tracks the 58-item list closely while saving under ten minutes of interview time. Collapsing the list into categories saves about as little and costs 21%, so it is the one option that is simply dominated.

The second is that a bad module damages its neighbours. The "usual month" module reports lower food than a 14-day recall and nearly doubles interview time, and its *non-food* answers run 25% above one comparison module and 32% above another, even though the non-food questions were word-for-word identical across every recall module. The cognitive load of a hard food module spills forward into whatever follows it. Questionnaire order is not a neutral administrative choice.

The third is that the bias is not a scalar you can subtract off. Recall modules under-report most for poorer households and for households with more adults, while apparent telescoping rises with asset wealth, so the error is correlated with exactly the things being studied. Squared poverty gaps for the collapsed-list and usual-month modules come in at more than twice the benchmark. Against all that, household consumption *rankings* are "remarkably stable" across modules, which is the one genuinely reassuring result in the paper, and the reason relative statements survive a change of instrument when absolute ones do not.

Their operational advice, which is unusually direct for a paper of this kind: if you must pick one recall module, use a long list over 7 days; never collapse the item list, because the savings are trivial against the bias; treat the subset list as an acceptable substitute if you can afford to lose the detail; drop "usual month" entirely; use infrequent supervision for household diaries unless literacy is low, where it "will dramatically underestimate consumption"; and pay for personal diaries only where the population is heavily urban, since they cost six to ten times a recall module. They also caution that the whole thing is setting-dependent. Rural Tanzania generalises to much of rural Africa and South and Southeast Asia; urban middle-income settings, much less so.

2.3.18 Recall periods actually in use

What the surveys in `LSMS_Library` actually do, which is a good deal less uniform than the literature's advice would suggest:

country	wave(s)	recall
Ethiopia	2011-12 … 2021-22 (5)	past one week
Togo	2018	7 days
Guatemala	2000	last 15 days
Panama	1997, 2003, 2008	last 15 days
Nigeria	2010Q3 … 2016Q1 (6)	7-day consumption
Nigeria	2018Q3, 2019Q1	7-day produced, 30-day purchase
Tanzania	2008-09 … 2020-21 (6)	consumed at home during…
Mali	2014-15	7 days
Serbia	2007	7-day diary
Burkina Faso	2014	7 days x 4 passages
Burkina Faso	2021-22	7 days and 30 days
GhanaLSS	1998-99, 2012-13, 2016-17	5 days x 6 asks
GhanaLSS	1991-92	2 days rural / 3 urban
GhanaLSS	1987-88, 1988-89	since my last visit

Note the two GhanaLSS entries at the bottom. "Since my last visit" is not a recall period at all, it is a recall period that varies by household and by enumerator schedule, and nothing in the delivered data records what it was. Comparing 1987-88 against 2016-17 therefore compares two instruments as much as two Ghanas, which is the point the list-length slide made earlier and the reason it is worth making twice.

2.3.19 Timing: what is a period?

For food, a day, probably: people schedule around meals. Surveys pick something longer, and each choice buys one error with another.

- A *diary* of what was eaten, meal by meal, is the ideal and the burden: hard to sustain, and paying for it buys a record, not an accurate one

- A *short* window is cheap but noisy: day-to-day variation in intake is not the error we are after

- *Usual* consumption asks the respondent to do our averaging for us, and to share our idea of "typical"; Beegle et al. found it fails outright

2.3.20 Inventories: purchases are not consumption

Storable staples — dry maize, rice, oil — are bought in bulk and drawn down. Perishables — fruit, fresh fish — are bought as eaten.

- A short window catches the fruit and misses the maize sack, or catches the sack and triples the week

- Averages out across households if purchase timing is unrelated to the visit; does not average out *within* one, which is where the line is applied

- Who buys in bulk is a matter of wealth: cash on hand, a dry store, a refrigerator. So the noise is on the rich, and on the staples

- Thirty days of visits (GLSS7) is one answer; asking what was *consumed*, as 8H does and 9B does not, is the other

2.3.21 Timing and inventories

The most natural "period" for food is probably a day; most people schedule around a certain number of meals per day. Some surveys implement a **food diary** method that tries to capture exactly what was eaten at different meals each day (either by individuals or by the household collectively). But the hassle of keeping up such a diary over an extended period is considerable, so it's difficult to induce respondents to keep an accurate record (paying them may tend to produce an *inaccurate* record).

What about a diary over a **short** period? Then the worry is that natural variation in food intake at such high frequencies will create a different form of measurement error, if what we're after is something like average intakes over time.

A handful of surveys try to have it both ways: they'll ask about what food was consumed in the most recent period (day/week/month), and then also ask about how much food is consumed in a *typical* period. This leans heavily on respondents *knowing* this, and also on having a shared understanding of what *typical* means.

Inventories are the part of this that the recall literature mostly skips. What a purchase module records is acquisition, and acquisition is consumption only for goods that cannot be kept. Fresh fish and fruit are bought as eaten; dry maize, rice and cooking oil are bought in quantities meant to last, and drawn down. Over a week, then, the perishables are measured about right and the staples are measured as either nothing or a month's worth, depending on whether the sack happened to be bought inside the window. If purchase timing is unrelated to the visit this averages out across households, which is why the population mean survives; it does not average out within a household, and it is the within-household number that gets compared with a poverty line. And the lumpiness is not randomly assigned. Buying in bulk takes cash on hand, somewhere dry to keep a sack, and for anything that spoils a refrigerator, all of which come with wealth; so richer households' food expenditure is measured with more noise than poorer households', and the noise sits on exactly the staples whose budget share Engel's law makes the marker of poverty. Two designs address it. GLSS7's six visits over thirty days lengthen the window until most sacks fall inside it, at the cost the previous slides describe. A consumption module asks what was eaten from stock as well as what was bought, at the cost of having to value what was drawn down, which is the own-production problem under another name. The object the aggregate is after is consumption; Section 8H asks for it, for own produce, and Section 9B does not, for purchases.

2.3.22 Price Indices

Two different price indices, meant to address two different problems:

- **Temporal**: prices rise between rounds. Use the CPI, or better, a survey-based Paasche index built from the survey's own unit values.
- **Spatial**: prices differ across regions *within* a round, often by more than they differ across a decade. A national poverty line applied to undeflated nominal consumption will find poverty wherever prices are high.

2.3.23 A household-specific Paasche index

Deaton & Zaidi recommend a household-specific Paasche index

$$P_i = \left[\sum_j s_{ij} \left(\frac{p_{ij}}{p_j^0} \right)^{-1} \right]^{-1},$$

with s_{ij} household i's budget share on item j.

2.3.24 Deflation

Spatial deflation is the step most often skipped and most often decisive. In a country the size of Ghana, the price of a fixed food basket can differ by 30% between Accra and the Upper East, which is comparable to the entire national increase in real consumption over a decade. An analysis that deflates over time but not over space will conclude that the north is poor for reasons that are partly an artefact of the accounting.

It also matters that the index is *household-specific*. A regional price index applies the same deflator to everyone in the region, and so implicitly assumes they consume the same bundle. They do not: the poor spend more of their budget on staples, so the relevant price index for the poor moves with staple prices. This is not a refinement; it is what makes the poverty rate respond correctly to a maize price shock.

2.3.25 Own production and the valuation problem

How much rides on this depends on where you are, and the range is wide. Deaton and Zaidi's Table 3.1 gives shares of the consumption aggregate:

Country	Food	of which purchased	of which home produced
Nepal	64.2	29.0	35.2
Ghana	65.2	44.4	20.8
Vietnam	50.9	34.1	16.8
Kyrgyz Rep.	44.5	33.4	11.1
Brazil	27.7	21.0	6.7
Panama	45.9	39.8	6.1
Ecuador	49.6	44.3	5.3
South Africa	30.4	28.2	2.2

So home production is a fifth of everything a Ghanaian household consumes and a third of its food, against a fiftieth in South Africa. Deaton puts the upper end higher still: in much of West Africa, he writes, "home production and hunting may account for a large share — perhaps more than a half — of total consumption" (ch. 1, p. 29). A quantity that large is not a correction to the aggregate; for these households it substantially *is* the aggregate, and it arrives entirely by imputation.

Which is worth stating the other way round: the more of the aggregate you impute, the less you are measuring and the more you are assuming.

Deaton and Zaidi make the point that imputed values are less variable than the truth would be, so leaning on them tends to *understate* inequality and poverty — the errors do not merely add noise, they push in a direction.

Deaton's example of imputation gone wrong is worth the retelling. A survey in West Africa recorded the water households fetched from local rivers and ponds, and priced every such item by the nearest traded equivalent. River water was therefore valued at the going price of Perrier in the nearest city, "thus endowing rural households with immense (but alas illusory) riches." Deaton and Zaidi return to it in a single dry clause: "water from the local pond is certainly different from L'Eau Perrier." The joke has a point — the algorithm was applied consistently, which is exactly what let it produce a consistent absurdity.

The unit-value hierarchy is worth dwelling on, because it is the first place in the workshop where a defensible answer requires a judgment rather than a formula. You have, for a given item, a set of reported (expenditure, quantity) pairs from households that bought it. Their ratio is a unit value, not a price: it embeds quality choice, package size, and reporting error, and a rich household's unit value for "rice" is higher partly because they are buying better rice.

The standard move is to take the median within the smallest cell with enough observations — cluster, then district, then region, then national — and use it for the households that did not buy. The median rather than the mean because unit values have a long right tail from misreported quantities. The hierarchy because the cluster cell will often be empty. And you should report how far up the hierarchy you had to climb, since an aggregate built mostly on national medians is a different object from one built on cluster medians.

2.3.26 Unit values, and what they hide

Look at the unit values directly, for one item, in one round.

```
uv = (acq.Expenditure / acq.Quantity).rename('unit_value')
uv = uv.replace([np.inf, -np.inf], np.nan).dropna()

item = uv.xs('2016-17', level='t').groupby('j').size().idxmax()
# commonest item
x = uv.xs('2016-17', level='t').xs(item, level='j')
print(item, ':', len(x), 'observations')
x.groupby('u').describe().round(2)          # by unit of measure
```

Two lessons, both visible in that table. Unit values can vary enormously within an item — ratios of 100 to 1 are common, and are misreported quantities, not real price dispersion. And they are only comparable *within* a unit of measure, which is why `u` is in the index.

```
# The median is the robust summary; the mean is not.
x.groupby('u').agg(['median', 'mean', 'count']).round(2).head()
```

2.3.27 Durables and housing

Durables: what enters consumption is the *flow of services*, not the purchase. For a stock S with service life L and interest rate r,

$$\text{use value} \;\approx\; S\left(r + \frac{1}{L}\right).$$

Housing: for owner-occupiers there is no rent, so impute it.

- hedonic regression of log rent on housing characteristics, among renters
- predict for owners
- fragile where the rental market is thin, which is most rural areas

2.4 Intra-Household Allocation

2.4.1 Adult equivalence

A household of six is not six times as well off as a household of one with the same total consumption. Two adjustments, usually conflated:

- **Household composition**: children cost less than adults — $\Rightarrow$ adult-equivalent scale $A_i = \sum_k a_{\text{age,sex}}$
- **Economies of scale**: shared goods, so cost rises less than proportionately — $\Rightarrow n_i^{\theta}$, with $\theta \in (0, 1)$

Combined: $\quad c_i = \dfrac{C_i}{A_i^{\theta}}.$

θ is not estimated in this session; it is *assumed*. Report the poverty rate for several values of it.

2.4.2 Barten: children re-price goods

A^θ makes a child a fraction of an adult *for everything.* Barten (1964): composition changes *which* goods a household needs,

$$\max_c\ U\left(\frac{c_1}{m_1(z)},\ldots,\frac{c_J}{m_J(z)}\right) \quad \text{s.t.} \quad \sum_j p_j c_j = x,$$

so a household with demographics z is the reference household at prices $p_j m_j(z)$. A child makes milk dear; substitution does the rest.

- the scale is a ratio of cost functions, $e(u, p \circ m(z))/e(u,p)$: it depends on prices, and in general on u
- A^θ is the case where every m_j is the same number: children change the *size* of the bundle, never its *shape*
- a shape that depends on z is not what A5 allows

2.4.3 Barten: children re-price goods

Barten's device is to put demographics into preferences good by good: effective consumption of good j is $c_j/m_j(z)$, so a household with children needs $m_j(z)$ units of j to get what the reference household gets from one. Since $\sum_j p_j c_j = \sum_j (p_j m_j)(c_j/m_j)$, the household solves the reference household's problem at prices $p \circ m(z)$, and every demographic effect on the bundle is a price effect: a child makes milk dear relative to beer, and the household substitutes accordingly. The equivalence scale, the ratio of what it costs the two households to reach the same u, is $e(u, p \circ m(z))/e(u,p)$. In general that depends on prices and on u; Lewbel's "independence of base" is the assumption that it does not depend on u, and it is what lets a scale be quoted as one number.

Now put this beside A5. Homotheticity says that at given prices every household consumes the same bundle up to scale, so a single price index $e(p)$ deflates everyone's expenditure. Barten says the shape of the bundle depends on z. The two are reconciled only by moving the homotheticity to *effective* consumption, U homothetic in $c/m(z)$. That buys independence of base, since the scale becomes $e(p \circ m(z))/e(p)$ with no u in it, but it costs the single deflator: the right index for a household of type z is $e(p \circ m(z))$, one per demographic type, and the scale is a ratio of price indices that moves when relative prices move. Cobb–Douglas is the one case where it does not, $e(p \circ m) = e(p) \prod_j m_j^{\beta_j}$; and in that case the m_j drop out of demand altogether, since $c_j = \beta_j x/p_j$ whatever z is, so the scale can be anything at all as far as the data are concerned. That is the general point in a special case. Barten scales are identified from how budget shares respond to the effective prices, so they are identified only to the extent that shares respond to prices at all; and under A5 shares do not respond to x, so the Engel and Rothbarth methods, which find the x at which two households have the same food share or the same adult-goods spending, have nothing to work with either. Pollak and Wales (1979) made the deeper point that demand data identify only *conditional* scales in any case, the cost of a child given that one has the child; a welfare comparison needs the unconditional one, and that has to come from an assumption, which is why θ is assumed rather than estimated on the previous slide.

2.5 Constructing Aggregate Welfare Measures

2.5.1 Poverty measures

The Foster–Greer–Thorbecke class: for poverty line z,

$$P_\alpha = \frac{1}{N} \sum_{i:c_i<z} \left(\frac{z - c_i}{z}\right)^\alpha .$$

- $\alpha = 0$: headcount ratio. Ignores *how* poor the poor are.
- $\alpha = 1$: poverty gap. The transfer needed, per capita, as a share of z.
- $\alpha = 2$: squared gap. Weights the poorest most; the only member of the family that satisfies the transfer axiom.

Weighted, always: $P_\alpha = \sum_i w_i \left(\frac{z-c_i}{z}\right)^\alpha / \sum_i w_i$.

2.5.2 Poverty, correctly weighted

```
sample = ghana.sample()

def fgt(c, w, z, alpha=0):
    """Weighted FGT_alpha.  c: consumption per adult equivalent; w: weights
    c, w = np.asarray(c, float), np.asarray(w, float)
    ok = np.isfinite(c) & np.isfinite(w)
    c, w = c[ok], w[ok]
    # Sum over the poor only.  Writing this as np.sum(w * gap**alpha) over
    # everybody looks equivalent, and is -- except at alpha=0, where
```

```python
    # 0**0 == 1 and every household in the country counts as poor.
    poor = c < z
    return np.sum(w[poor] * ((z - c[poor]) / z) ** alpha) / np.sum(w)

wave = '2016-17'
c = food_total.xs(wave, level='t')
w = sample.xs(wave, level='t').weight.reindex(c.index)

z = c.quantile(0.25)          # a placeholder line; see the exercise
for a in (0, 1, 2):
    print(f"P_{a} = {fgt(c, w, z, a):.4f}")

# The line is an unweighted quartile, so the UNWEIGHTED headcount must come
# back as 0.25 by construction.  It is the one number here we know in
# advance, which makes it the right thing to check the code against.
print(f"unweighted P_0 = {fgt(c, np.ones(len(c)), z, 0):.4f}")
print(f"mean weight, poor = {w[c < z].mean():.3f}"
      f"   non-poor = {w[c >= z].mean():.3f}")
```

The unweighted headcount returns 0.2500 — as it must be, since the line *is* the unweighted lower quartile. That is the check: it is the one quantity here whose value you know before running the code, so it is the one that tells you whether the code is right. If it comes back as 1.0000, you have found the 0^0 trap.

The weighted headcount is 0.2152 — three and a half points lower. The reason is in the line below it: poor households carry a mean weight of 0.86 against 1.05 for everyone else, because the design over-sampled them. You may recall that's Session 1's lesson.

It is worth knowing what these same lines said when `food_total` was purchases only: a weighted headcount of 0.154 against the same 0.25, and mean weights of 0.62 for the poor against 1.13. The gap has narrowed because the households a purchases-only measure put in the bottom quartile were, to a large extent, the ones that grow their own food, and those are also the ones the design over-sampled. Valued, their own production moves them up. On purchases alone, "poor" had largely meant "does not buy".

None of which rescues the line itself. A quartile of the distribution is not a poverty line: it fixes the unweighted headcount at 0.25 by construction, in every country and every year, which is a fact about arithmetic rather than about Ghana. A real line is absolute — the cost of a fixed bundle — and does not move with the distribution, and does not move when everyone gets poorer. Fixing that is the first exercise.

2.5.3 Inequality

- **Lorenz curve**: cumulative share of consumption against cumulative share of population. Everything else is a summary of it.
- **Gini**: twice the area between the Lorenz curve and the 45° line. Bounded, familiar, and insensitive where it matters most.
- **Generalized entropy** $GE(\alpha)$: decomposable into within- and between-group parts. $GE(0)$ is mean log deviation, $GE(1)$ is Theil.

2.5.4 Inequality: Atkinson

The opening slide's $I = U\big(\int w\,dF\big) - \int U(w)\,dF$ is in utils, which nobody reports. Let w_{ede} solve $U(w_{\text{ede}}) = \int U\,dF$: the level which, given to everyone, yields the same total utility as the actual spread. Then $I = U(\bar{w}) - U(w_{\text{ede}})$, and

$$A = 1 - \frac{w_{\text{ede}}}{\bar{w}}$$

is the same judgement in cedis per cedi, with $U(w) = w^{1-\varepsilon}/(1-\varepsilon)$ and ε assumed, like θ.

$A = 0.3$: this society would give up 30% of mean consumption to have the rest shared equally. Prefer the Lorenz curve to any scalar.

2.5.5 Inequality

The Atkinson index is the opening slide's I with the units fixed. That I is the gap between the utility of the mean and mean utility, which is a quantity in utils, and utils are not a currency anyone reports. Solving for the consumption level that delivers the same mean utility converts the gap into cedis, and dividing by $\bar{w}$ makes it unit free. Same value judgement, expressed as a number a finance ministry can act on.

The judgement is ε, and it is worth being blunt that it is assumed rather than estimated, exactly as θ was in the equivalence scale. Taking $U(w) = w^{1-\varepsilon}/(1-\varepsilon)$,

$$w_{\text{ede}} = \left[\int w^{1-\varepsilon} dF(w)\right]^{1/(1-\varepsilon)},$$

with the limiting case $\varepsilon = 1$ giving the geometric mean, $w_{\text{ede}} = \exp \int \log w\,dF(w)$. At $\varepsilon = 0$ nobody cares about distribution and $A = 0$ whatever the data look like; as ε grows the index attends more and more to the bottom of the distribution, and in the limit only the poorest household counts. Report it for several values, and let the reader find their own.

Two things follow that are easy to miss. The Atkinson family and the generalized entropy family are the same orderings wearing different clothes: $A(\varepsilon)$ is a monotone transform of $GE(1-\varepsilon)$, so choosing between them is a choice about presentation, not about ethics. And because A is built from a concave U, it is guaranteed to respect the Pigou–Dalton principle, which the Gini also does but which a variance of logs, still common in the applied literature, does not.

2.5.6 Lorenz and Gini

```
def lorenz(c, w):
    c, w = np.asarray(c, float), np.asarray(w, float)
    ok = np.isfinite(c) & np.isfinite(w) & (c >= 0)
    c, w = c[ok], w[ok]
    order = np.argsort(c)
    c, w = c[order], w[order]
    p = np.cumsum(w) / np.sum(w)
    L = np.cumsum(w * c) / np.sum(w * c)
    return np.concatenate([[0], p]), np.concatenate([[0], L])

def gini(c, w):
    p, L = lorenz(c, w)
    return 1 - 2 * np.trapezoid(L, p)

print(f"Gini (food, purchased and own-produced, {wave}) = {gini(c, w):.3f}"
```

The Atkinson index needs one more thing than the Gini does, and the code should make you say it out loud: how averse to inequality you are.

```
def atkinson(c, w, eps):
    """Weighted Atkinson index. eps > 0 is inequality aversion."""
    c, w = np.asarray(c, float), np.asarray(w, float)
    ok = np.isfinite(c) & np.isfinite(w) & (c > 0)
# log/power need c > 0
    c, w = c[ok], w[ok]
    mean = np.sum(w * c) / np.sum(w)
    if np.isclose(eps, 1):
        ede = np.exp(np.sum(w * np.log(c)) / np.sum(w))
```

```
    else:
        ede = (np.sum(w * c ** (1 - eps)) / np.sum(w)) ** (1 / (1 - eps))
    return 1 - ede / mean

for eps in (0.5, 1.0, 2.0):
    print(f"Atkinson A({eps}) = {atkinson(c, w, eps):.3f}")
```

The library draws the curve too, and does one thing the hand-rolled plot does not: it writes every choice that moves the curve into the subtitle.

```
import matplotlib.pyplot as plt

waves = ['1998-99', '2005-06', '2012-13', '2016-17']
ll.visualizations.lorenz_curve(ghana, waves, per='person', basis='total')
plt.show()
```

The Gini in the legend is not the one you computed above, and the subtitle says why: it counts *persons*, weighting each household by its size, where the code above counted households. Both are Ginis of food consumption in the same year. Neither is wrong; the one you report is a choice, and the chart makes you state it.

If two Lorenz curves cross, the ranking of the two distributions depends on which inequality measure you chose, and no scalar will rescue you. Look before you summarize.

2.5.7 One index, or several?

Everything here divides nominal expenditure by a scalar: a Paasche index over space, a CPI over time, an equivalence scale over composition. Three scalars, one household, one number out.

- one deflator is exact at **rank one**: budget shares do not vary with total expenditure
- but they do — Engel's law, which opened this session
- a line *and* a deflator are exact at **rank two** (Gorman), which Engel's law permits
- so on what grounds is a two-index procedure adequate?

Hold the question: session 3 answers it for Uganda.

2.5.8 Final thoughts

It would be dishonest to teach the Deaton–Zaidi procedure as best practice and leave it there, because the profession's own best estimates say the procedure is not adequate for the data we are using it on. It is also the procedure every statistical agency uses, it is what your referees will expect, and you must be able to execute it competently — which is why it comes first and in full.

The order is deliberate: learn the standard method properly here, learn in session 3 what it costs, and then decide, case by case, whether the cost matters for the question in front of you. That is a different thing from never having known.

2.5.9 Exercises

1. The poverty line used above is fake. Build a real one: take the food bundle consumed by households in the second and third deciles, price it at national median unit values, and scale it to 2,900 kcal per adult equivalent. Recompute P_0, P_1, P_2. Notebook: `poverty_line.ipynb`

2. Deflate spatially. Construct a regional Paasche index from the survey's own unit values, apply it, and report how much of the north–south poverty gradient survives. Notebook: `spatial_deflation.ipynb`

3. Vary θ in $c_i = C_i/A_i^\theta$ over $[0.5, 1.0]$ and plot the headcount ratio against it. Over what range does the *ranking* of regions change? Notebook: `equivalence_scales.ipynb`

3 Demand and Welfare

Where household surveys earn their keep for policy: what happens to whom when a price moves.

3.1 Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton, chs. 4 (§4.2 on Engel curves) and 5 (all)

- Deaton & Muellbauer (1980), "An Almost Ideal Demand System," *AER* 70:312–326

- Lewbel (1991), "The rank of demand systems," *Econometrica* 59:711–730
- Ligon, "Household Welfare from Disaggregate Demands" (working paper) — the CFE system we estimate today

3.2 Budget shares and Engel curves

Let $w_{ij} = p_j q_{ij}/x_i$ be household i's budget share on good j, x_i total expenditure. The Working–Leser form,

$$w_{ij} = \alpha_j + \beta_j \log x_i + \gamma_j' z_i + u_{ij},$$

fits food shares remarkably well across countries and decades.

- $\beta_{\text{food}} < 0$ is Engel's law
- z_i: household size and composition, region, season
- Estimated by OLS, weighted, clustered on the EA

3.3 Engel's law as a diagnostic

The food share is the oldest welfare indicator in economics and still one of the better ones, precisely because it requires no price data and no deflation. If your constructed consumption aggregate does not produce a declining food share in log total expenditure, something is wrong with the aggregate, and you should find out what before going further. It is the cheapest specification check available, and it catches a surprising share of the errors that arise in session 2 — a mis-scaled own-production imputation, in particular, shows up immediately as a food share that is flat or rising.

3.4 Equivalence scales from behaviour

Session 2 *assumed* a scale. Demand analysis lets you estimate one.

- **Engel method**: two households are equally well off if they have the same food share. Scale = the expenditure ratio that equalizes it.
- **Rothbarth method**: use expenditure on adult goods (alcohol, tobacco, adult clothing) as the indicator instead.
- The two disagree, systematically and by a lot. Deaton ch. 4 explains why: the Engel method conflates the cost of a child with the child's effect on the household's needs for food specifically.

3.5 Prices, and where to get them

LSMS surveys rarely carry a price questionnaire. They carry unit values, and unit values are contaminated by quality choice:

$$\log v_{ij} = \log p_j + \underbrace{\psi_j \log x_i}_{\text{quality}} + \epsilon_{ij}.$$

Deaton's solution exploits the design: *within a cluster*, all households face the same price. So

- regress $\log v_{ij}$ on $\log x_i$ and household characteristics *with cluster fixed effects* to strip out quality;
- the cluster effects are the price variation you wanted.

This is the single best argument for caring about `v` in the index.

3.6 The rank of a demand system

Stack the Engel curves. The *rank* of a demand system (Lewbel 1991) is the dimension of the space spanned by the budget shares viewed as functions of total expenditure.

- rank 1: shares do not vary with x — homothetic
- rank 2: Working–Leser, AIDS
- rank 3: QUAIDS; Gorman's bound for exactly aggregable systems

Why you should care: **if the rank exceeds one, no single price index can deflate nominal expenditure so as to keep it monotone in utility when relative prices move.** Rank r needs r indices.

3.7 The rank of a demand system

This is the result that indicts session 2, and it is worth being precise about the indictment. Everything we did there — the Deaton–Zaidi household-specific Paasche index, the spatial deflator, the temporal one — constructs a *single* scalar per household and divides by it. That procedure is exactly right if the demand system has rank one, defensible as an approximation at rank two, and simply not available at higher rank: there is no scalar function of prices whose quotient with nominal expenditure orders households the way utility does.

Estimates for Uganda put the rank at three at least, with a best estimate of six. So the deflation in session 2 is not a refinement we skipped for lack of time; it is a procedure that cannot be made correct by being more careful with the index. What follows is the alternative: estimate the welfare number directly from behaviour, and never construct a price index at all.

3.8 Constant Frisch Elasticity demands

Let λ_i be household i's marginal utility of expenditure. Frischian (λ-constant) demands take the CFE form

$$f_j(p,\lambda) = \left(\frac{\alpha_j}{\lambda p_j}\right)^{\beta_j}, \qquad U(c) = \sum_j \alpha_j \beta_j \frac{c_j^{1-1/\beta_j} - 1}{\beta_j - 1}.$$

- β_j is the **Frisch elasticity** of demand for good j: necessities small, luxuries large
- $\beta_j = \beta$ for all j gives CES; $\beta \to 1$ gives Cobb–Douglas
- unrestricted rank, so it does not impose what we just said is false

3.9 The estimating equation

Take logs of expenditure on good j. Writing $w_i \equiv -\log \lambda_i$,

$$\log x_{it}^j \;=\; a_t^j \;+\; \beta_j\, w_{it} \;+\; \gamma_j' d_{it} \;+\; \varepsilon_{it}^j.$$

This is a **one-factor model**. The $(it) \times j$ matrix of log expenditures, once good-time effects a_t^j and characteristics d_{it} are swept out, is rank one: loadings β, factor w.

- estimate by SVD of the residual matrix, with missing data
- d_{it} enters as good-specific Barten scales, not "independence of base"
- needs **no prices** and **no total expenditure** — only expenditures on a subset of goods

3.10 $w = -\log \lambda$ as a welfare measure

λ is the marginal utility of expenditure, so it *falls* as a household gets better off. Hence $w = -\log \lambda$ is increasing in welfare.

- comparable across households facing different prices — the prices are in a_t^j, not in w
- comparable across household compositions — composition is in $\gamma_j' d$
- derived from behaviour, not from an assumed equivalence scale or an assumed price index
- w is the partial derivative of indirect utility with respect to total expenditure: a genuine welfare object, though not itself a utility level

3.11 $w = -\log\lambda$ as a welfare measure

It is worth being clear about what this does and does not deliver, because the claim is easy to overstate in both directions.

What it delivers: a scalar per household-period that is comparable across households, estimated from expenditures on a subset of goods, with no price data and no total-expenditure measure. That is a remarkable amount to get from very little, and it is precisely the situation an LSMS analyst is in.

What it does not deliver: a utility level. One cannot in general integrate back from λ to the indirect utility function — that would require demands for *all* goods. So w supports statements about who is better off than whom, and about how a distribution shifts; it does not support a claim to have measured utility. Much of the appeal is exactly that the weaker claim is enough for the questions we actually ask of these data.

3.12 Welfare consequences of a price change

For a small change in the price of good j, the compensating variation is approximately

$$\mathrm{CV}_i \;\approx\; q_{ij}\,\Delta p_j \;=\; w_{ij} x_i \frac{\Delta p_j}{p_j}.$$

- to first order you need only the *budget share* — no elasticities
- the distributional incidence follows from how w_{ij} varies with x_i, which is the Engel curve you already estimated
- second-order terms need the demand system, and are usually small relative to the first-order term for modest price changes

3.13 Welfare consequences of a price change

This is the payoff for the whole exercise, and it is worth being explicit about how little machinery the first-order answer requires. A finance ministry contemplating the removal of a fuel subsidy does not need a structural demand system to know the first-order incidence: it needs the budget share of fuel by decile, which is one `groupby` on a well-constructed consumption aggregate. The demand system buys you the behavioural response — how much households substitute away — which matters for the revenue calculation and for large changes, and much less for the distributional one.

The order of operations in policy work therefore usually runs backwards from the way it is taught. Get the aggregate right, get the shares right, report the first-order incidence with standard errors, and only then argue about elasticities.

3.14 Budget shares

```
# Show tracebacks without the library's internal frames: the line that
# failed, and why.  Change Plain to Verbose if you ever want the rest.
%xmode Plain

# The first call that builds a table may print "DVC unavailable ...
# falling back to manual aggregation".  That is about how the library
# fetches its raw files, not about your data; the numbers are the same,
# and it does not recur once the table is built.

# These notebooks are read two ways: on your own screen, and projected onto
# a wall in a lit room where thin lines and pale markers are simply not
# there.  Those want different plots, so the difference is a switch rather
# than a compromise that suits neither.  Leave it False; the lecturer flips
# it.  Everything below the 'if' is about the wall, not about your laptop.
PROJECTOR = False

import matplotlib as mpl
mpl.rcParams.update({'figure.dpi': 110, 'font.size': 11,
                     'lines.linewidth': 1.8})
if PROJECTOR:
    mpl.rcParams.update({
        'figure.figsize': (7, 4.5),
        'font.size': 15, 'axes.labelsize': 15, 'axes.titlesize': 17,
```

```
        'xtick.labelsize': 13, 'ytick.labelsize': 13, 'legend.fontsize': 13
        'lines.linewidth': 3.0, 'lines.markersize': 9, 'axes.linewidth': 1.
    })

import lsms_library as ll
import numpy as np, pandas as pd

ghana = ll.Country('GhanaLSS')
food = ghana.food_expenditures()
sample = ghana.sample()

wave = '2016-17'
# food_expenditures is indexed (i, t, v, j, s).  Collapse to one row per
# household-item before doing anything else: unstacking the raw index
# would ask pandas for a 13918 x 373678 array.
f = food.xs(wave, level='t').squeeze().groupby(['i', 'j']).sum()
x = f.groupby('i').sum()                        # total food expenditure
shares = f.div(x, level='i').unstack('j').fillna(0.0)
shares.shape

# The ten largest items by mean budget share
top = shares.mean().sort_values(ascending=False).head(10)
top.round(4)
```

3.15 An Engel curve

```
import statsmodels.api as sm

# household_characteristics is a count of people by age-sex cell, indexed
# (t, v, i).  Sum across cells for household size, then drop to i alone so
# it aligns with the shares.
chars = ghana.household_characteristics().xs(wave, level='t')
hhsize = chars.sum(axis=1).groupby('i').first().rename('hhsize')

s1 = sample.xs(wave, level='t')
item = top.index[0]

d = pd.concat([shares[item].rename('w'), x.rename('x'), hhsize,
               s1.v.rename('v'), s1.weight.rename('wt')], axis=1).dropna()
```

```python
d = d[(d.x > 0) & (d.hhsize > 0)]
d['logx'] = np.log(d.x.astype(float))
d['logn'] = np.log(d.hhsize.astype(float))

res = sm.WLS(d.w.astype(float),
             sm.add_constant(d[['logx', 'logn']].astype(float)),
             weights=d.wt.astype(float)
             ).fit(cov_type='cluster', cov_kwds={'groups': d.v.astype(str)}
print(res.summary().tables[1])
```

Note `cov_type`'cluster'. `Session 1 established why: without it the standard errors on =logx` are too small by roughly $\sqrt{\text{deff}}$.

```python
import matplotlib.pyplot as plt

# Nonparametric Engel curve: weighted mean share by expenditure ventile.
# A weighted mean is a ratio of two sums, so no groupby-apply is needed.
d['bin'] = pd.qcut(d.logx, 20, labels=False, duplicates='drop')
curve = (d.w * d.wt).groupby(d.bin).sum() / d.wt.groupby(d.bin).sum()
mid = d.logx.groupby(d.bin).mean()

fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(mid, curve, 'o-')
ax.set_xlabel(r'$\log x$ (total food expenditure)')
ax.set_ylabel(f'budget share, {item}')
plt.show()
```

3.16 Stripping quality out of unit values

```python
acq = ghana.food_acquired().xs(wave, level='t')
uv = np.log((acq.Expenditure / acq.Quantity).replace([np.inf, -np.inf], np.

j0 = uv.groupby('j').size().idxmax()
u0 = uv.xs(j0, level='j').groupby('i').mean().rename('logv')

e = pd.concat([u0, d.logx, d.v], axis=1).dropna()

# Deaton's within-cluster estimator. There are ~1000 clusters, so do not
# build the dummy matrix; by Frisch-Waugh-Lovell, demeaning within cluster
# gives the identical coefficient at a fraction of the cost.
```

```
for col in ('logv', 'logx'):
    e[col + '_d'] = e[col].astype(float) - e.groupby('v')[col].transform('m

within = sm.OLS(e.logv_d, e[['logx_d']]).fit(
    cov_type='cluster', cov_kwds={'groups': e.v.astype(str)})
print(f"quality␣elasticity␣psi␣=␣{within.params['logx_d']:.3f}"
      f"␣␣␣(se␣{within.bse['logx_d']:.3f})")
```

A ψ near zero says households of different means pay the same for this item — it is a homogeneous good. A large ψ says "rice" in the questionnaire is several different goods in the market.

3.17 Estimating a CFE system

The CFE estimator needs several waves and a food classification that is consistent across them. Ghana's GLSS has the waves but no harmonized food aggregation; Uganda has both, and it is the panel we use for the rest of the workshop. So we switch countries here, deliberately.

```
import lsms_library as ll
from cfe import Regression
import numpy as np, pandas as pd

uga = ll.Country('Uganda')
x = uga.food_expenditures().squeeze()

# Collapse the survey's fine item codes onto the harmonized aggregate
# labels: 'Matoke (bunch)', 'Matoke (heap)', ... all become 'Matoke'.
agg = (uga.categorical_mapping['harmonize_food']
         .set_index('Preferred␣Label')['Aggregate␣Label'].to_dict())
x = x.rename(index=agg, level='j')
x = x.groupby(x.index.names).sum()
print(x.index.get_level_values('j').nunique(), 'goods␣after␣aggregation')
```

`cfe` insists on the index being exactly (i, t, m, j) — household, period, *market*, good. A market is a set of households facing common prices, so the enumeration area is the natural choice; we start with a single market, which puts all the price variation into the good-time effects a_t^j.

```
y = np.log(x.replace(0, np.nan).dropna()).groupby(['i', 't', 'j']).sum()
y = pd.concat({1: y}, names=['m']).reorder_levels(['i', 't', 'm', 'j']).sor
```

```
d0 = uga.household_characteristics()
d = d0.assign(
     Girls=d0[[f'F {a}' for a in ['00-03', '04-08', '09-13', '14-18']]].sum(
     Boys =d0[[f'M {a}' for a in ['00-03', '04-08', '09-13', '14-18']]].sum(
     Women=d0[[f'F {a}' for a in ['19-30', '31-50', '51+']]].sum(axis=1),
     Men  =d0[[f'M {a}' for a in ['19-30', '31-50', '51+']]].sum(axis=1),
)[['Girls', 'Boys', 'Women', 'Men', 'log HSize']].dropna(how='any')
d = d.groupby(['i', 't']).first()
d = pd.concat({1: d}, names=['m']).reorder_levels(['i', 't', 'm']).sort_ind

r = Regression(y=y, d=d)
beta = r.get_beta()          # a few seconds
print(len(beta), 'goods estimated')
```

3.18 Frisch elasticities

```
print("least elastic:"); print(beta.sort_values().head(6).round(2).to_strin
print("\nmost elastic:"); print(beta.sort_values().tail(6).round(2).to_stri
```

Read those two lists before reading anything else. The ordering is the model's main testable implication and it is not imposed: nothing in the estimator knows which goods are staples. If salt and cassava do not come out at the bottom and fruit and coffee at the top, something is wrong with the data or with the aggregation.

```
ax = r.graph_beta(xlabel=r'Estimates of $\beta$')
```

We did not estimate demands for all seventy-six goods. Items too few households ever report cannot support a coefficient, and the estimator drops them without being asked. Report how many survived, because the answer depends on the aggregation you chose.

Household composition enters through γ, one coefficient per good per characteristic — good-specific Barten scales rather than a single equivalence scale imposed on everything:

```
r.get_gamma().round(2).head(8)
```

Before trusting any of it, look at the fit:

```
import matplotlib.pyplot as plt

fit = pd.DataFrame({'actual': r.y,
                    'predicted': r.get_predicted_log_expenditures()}).dropn
```

```
ax = fit.plot.scatter(x='predicted', y='actual', figsize=(5, 5),
                      s=8 if PROJECTOR else 2,
                      alpha=0.35 if PROJECTOR else 0.2)
lim = [fit.min().min(), fit.max().max()]
ax.plot(lim, lim, 'k--', lw=1.5)
ax.set_title(f"log expenditures, R^2 = {fit.corr().iloc[0, 1] ** 2:.2f}")
plt.show()
```

Now the welfare measure itself. This one takes about half a minute.

```
w = r.get_w()                    # w = -log lambda, indexed (i, t, m)
w.groupby('t').describe()[['count', 'mean', 'std']].round(3)
```

Estimating this took most of a minute, and sessions 4 and 5 need the same object. Save it rather than refitting: the pickle carries β, γ, w, and the predicted expenditures together.

```
from pathlib import Path

r.predicted_expenditures()               # populate before saving
out = Path.home() / '.cache' / 'hhsurveys' / 'uganda.rgsn'
out.parent.mkdir(parents=True, exist_ok=True)
r.to_pickle(str(out))

# and to get it back:
#   import cfe
#   r = cfe.read_pickle(str(out))
print('saved', out)
```

3.19 The Engel pie

An Engel curve shows one good at a time. With a whole system estimated we can show all of them at once: draw the predicted budget *composition* as a pie, and let the radius be $\log x$. Poor households at the centre, rich at the rim.

```
import matplotlib.pyplot as plt
from matplotlib import cm

xhat = r.predicted_expenditures()
xbar = xhat.groupby(['i', 't', 'm']).sum()
p_j = ((r.y.unstack('j') > 0) + 0.).mean()        # prob. good j is purchased
```

```
lo, hi = xbar.quantile(0.05), xbar.quantile(0.95)
Y = np.geomspace(lo, hi, 60)

fig, ax = plt.subplots(figsize=(7.5, 6))
wedges = None
for k in range(len(Y) - 1, 0, -1):
    ax.set_prop_cycle('color', cm.tab20.colors)
    shares = r.expenditures(Y[k]) * p_j
    out = ax.pie(shares, radius=np.log(Y[k]) / np.log(hi), counterclock=Fal
    if wedges is None:                          # keep the outermost ring
        wedges, goods = out[0], shares.index.tolist()

ax.arrow(0, 0, 1, 0, shape='full', head_width=.04, length_includes_head=Tru
ax.annotate(r'$\log␣x$', xy=(1, 0), color='red', va='center_baseline')

# 41 goods will not fit in a legend; name the ten largest at the rim.
big = np.argsort(r.expenditures(Y[-1]).to_numpy() * p_j.to_numpy())[::-1][:
ax.legend([wedges[b] for b in big], [goods[b] for b in big],
          loc='center␣left', bbox_to_anchor=(1.02, 0.5),
          frameon=False, fontsize=8, title='largest␣ten,␣at␣the␣rim')
plt.show()
```

Wedges that widen as you move outward are luxuries; wedges that pinch are necessities. This is Engel's law, for forty-one goods simultaneously, and it is the same information as the β plot in a form you can show to someone who does not know what a Frisch elasticity is.

3.20 Exercises

1. Estimate the Working–Leser food share equation on the *full* consumption aggregate rather than food alone, and test whether β differs between urban and rural households.

2. Implement the Engel equivalence scale: find the expenditure ratio that equates the predicted food share of a household with two adults and two children to that of a household with two adults. Compare with the θ-scale you used in session 2.

3. Compute the first-order welfare cost of a 20% rise in the price of the largest food item, by consumption decile. Who loses most, in cedis and as a share of expenditure? Are those the same households?

4. Re-estimate the CFE system using the enumeration area as the market m rather than a single national market. How much do the β_j move? What have you assumed away by using one market?

5. Regress w on log total food expenditure. The two are measuring the same thing in principle; how closely do they agree, and which households do they disagree about?

4 From Cross-Sections to Dynamics

Two ways to get change out of survey data: build a panel from cohorts, or use one that already exists.

4.1 Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton (1985), "Panel data from time series of cross-sections"
- Deaton, ch. 2 §2.7 and ch. 6
- Abadie, Athey, Imbens & Wooldridge (2023), "When should you adjust standard errors for clustering?" *QJE* 138:1–35

4.2 The problem

Ghana's GLSS gives seven rounds, but almost never the same household twice. We want

$$\Delta \log c_{it} = \alpha_i + \delta_t + \beta' \Delta z_{it} + \epsilon_{it},$$

and α_i is not identified because i appears once.

Deaton's answer: give up on households and follow *cohorts*. A cohort — say, everyone born 1965–69 — is present in every round, and its membership is fixed by construction.

4.3 Pseudo-panels

Define cohort c by birth year of the head. For each (c, t) compute the weighted mean $\bar{y}_{ct}$. Then

$$\bar{y}_{ct} = \alpha_c + \delta_t + \beta' \bar{z}_{ct} + \bar{\epsilon}_{ct}$$

is a genuine panel, in $C \times T$ cells, and α_c is estimable.

The catch: $\bar{y}_{ct}$ is a *sample* mean, so it carries measurement error of order $1/\sqrt{n_{ct}}$. With $\bar{z}$ on the right-hand side that is classical errors-in-variables, and $\hat{\beta}$ attenuates.

4.4 Cohort size is the design decision

Everything about a pseudo-panel turns on one tradeoff, and it is worth stating it as a tradeoff rather than as a rule of thumb. Narrow cohorts (five birth years) give you many cells, so many degrees of freedom, but few households per cell, so noisy cell means and severe attenuation. Wide cohorts (fifteen birth years) give precise cell means but few cells and a cohort that is internally heterogeneous, so the fixed effect α_c is absorbing variation you might have wanted.

Deaton's own guidance is that cells want at least a hundred observations, and Verbeek and Nijman later showed that the errors-in-variables correction is straightforward once you have the within-cell variances, which the survey gives you for free. The practical advice is: choose the cohort width, report the cell counts, and show that the answer does not change much at the next width up. An analysis that does not report n_{ct} is hiding the one number a reader needs.

4.5 Identification: age, cohort, and time

You cannot have all three. Since

$$\text{age} = \text{year} - \text{birth year},$$

age, period, and cohort effects are exactly collinear. Something must be restricted:

- drop period effects (a "pure lifecycle" reading), or
- restrict period effects to sum to zero and be orthogonal to trend (Deaton & Paxson's normalization), or
- bring in outside information on one of the three

This is not a technical nuisance to be worked around. It is a statement that the data cannot answer the question without an assumption, and the assumption should be argued for.

4.6 A genuine panel: Uganda

The Uganda National Panel Survey follows households from 2005–06 to 2018–19. With the same i in multiple t:

$$y_{it} = \alpha_i + \delta_t + \beta' z_{it} + \epsilon_{it}$$

is estimable by within transformation.

- α_i absorbs everything time-invariant — soil quality, ability, the district's road
- what remains identified is the effect of variation *within* the household over time
- and β is now a within estimate, which is a different parameter from the cross-sectional one, not a better-identified version of it

4.7 Inference

Cluster at the level of treatment assignment, not at the level of sampling.

- sampling clusters (EAs): cluster the SEs, per session 1
- if the variation you exploit is at district level, cluster on district
- few clusters ($G < 40$) $\Rightarrow$ wild cluster bootstrap
- two-way (household and time) when T is not small

Abadie et al. (2023): clustering is about *design* or *sampling*, and if neither applies, clustering is not conservative — it is wrong.

4.8 Building a pseudo-panel

```
# Show tracebacks without the library's internal frames: the line that
# failed, and why.  Change Plain to Verbose if you ever want the rest.
%xmode Plain

# The first call that builds a table may print "DVC unavailable ...
# falling back to manual aggregation".  That is about how the library
# fetches its raw files, not about your data; the numbers are the same,
# and it does not recur once the table is built.
```

```
# These notebooks are read two ways: on your own screen, and projected onto
# a wall in a lit room where thin lines and pale markers are simply not
# there.  Those want different plots, so the difference is a switch rather
# than a compromise that suits neither.  Leave it False; the lecturer flips
# it.  Everything below the `if' is about the wall, not about your laptop.
PROJECTOR = False

import matplotlib as mpl
mpl.rcParams.update({'figure.dpi': 110, 'font.size': 11,
                     'lines.linewidth': 1.8})
if PROJECTOR:
    mpl.rcParams.update({
        'figure.figsize': (7, 4.5),
        'font.size': 15, 'axes.labelsize': 15, 'axes.titlesize': 17,
        'xtick.labelsize': 13, 'ytick.labelsize': 13, 'legend.fontsize': 13
        'lines.linewidth': 3.0, 'lines.markersize': 9, 'axes.linewidth': 1.
    })

import lsms_library as ll
import numpy as np, pandas as pd

ghana = ll.Country('GhanaLSS')
roster = ghana.household_roster()
sample = ghana.sample()
food = ghana.food_expenditures()

# Age of the head, by (t, i).  Relationship == 'Head' identifies the head.
heads = roster[roster.Relationship.str.strip().str.lower().eq('head')]
age = heads.groupby(['t', 'i']).Age.first().rename('age')
age.groupby('t').describe().round(1)

# Wave midpoint year, so we can turn age into a birth year
year = {w: int(w[:4]) + 1 for w in ghana.waves}

x = food.groupby(['t', 'i']).sum().squeeze().rename('c')
d = pd.concat([x, age], axis=1).dropna()
d['year'] = [year[t] for t in d.index.get_level_values('t')]
d['born'] = d.year - d.age
d = d[(d.age >= 25) & (d.age <= 70)]
```

```
# Five-year cohorts
d['cohort'] = (d.born // 5) * 5
d['logc'] = np.log(d.c.where(d.c > 0))

# `sample` is indexed (i, t); everything else here is (t, i).
reindex
# against a differently-ordered MultiIndex does not raise -- it returns all
# NaN, and the pseudo-panel below comes out empty with no error anywhere.
# Reorder explicitly, then check that the merge actually matched.
w_all = sample.weight.reorder_levels(['t', 'i']).sort_index()
d['w'] = w_all.reindex(d.index)
assert d.w.notna().mean() > 0.9, f"only {d.w.notna().mean():.1%} of weights


# Weighted cell means, as ratios of sums.  groupby-apply returning a Series
# is fragile across pandas versions; this is not.
dd = d.dropna(subset=['logc', 'w']).reset_index()
grp = dd.groupby(['cohort', 't'])
cells = pd.DataFrame({
    'logc': (dd.w * dd.logc).groupby([dd.cohort, dd.t]).sum() / grp.w.sum()
    'age':  (dd.w * dd.age).groupby([dd.cohort, dd.t]).sum() / grp.w.sum(),
    'n':    grp.size(),
})
cells.head(12)

# Deaton's rule of thumb: cells want n >= 100.  How many survive?
print(f"{len(cells)} cells; {(cells.n >= 100).mean():.3f} of them have n >=
cells.n.describe().round(0)
```

4.9 Lifecycle consumption

```
import matplotlib.pyplot as plt

big = cells[cells.n >= 100].reset_index()
fig, ax = plt.subplots(figsize=(6.5, 4))
for c, g in big.groupby('cohort'):
    if len(g) >= 3:
        ax.plot(g.age, g.logc, 'o-', label=f"born {int(c)}-{int(c)+4}")
```

```
ax.set_xlabel('age of head')
ax.set_ylabel('mean log food consumption')
ax.legend(frameon=False, fontsize=7, ncol=2)
plt.show()
```

Each line is one cohort tracked across rounds. Where the lines lie on top of one another, there is no cohort effect and the shape is lifecycle. Where they are shifted vertically, later cohorts are better off at every age — a cohort effect. Where *every* line jumps in the same round, that is a period effect, or a change in the questionnaire.

4.10 The Uganda panel

```
uga = ll.Country('Uganda')
print(uga.waves)
print(uga.data_scheme)

import statsmodels.api as sm

ufood = uga.food_expenditures()
uc = ufood.groupby(['t', 'i']).sum().squeeze()
uc = np.log(uc.where(uc > 0)).rename('logc')

un = uga.household_characteristics().sum(axis=1).groupby(['t', 'i']).first(
un = np.log(un.astype(float).where(un > 0)).rename('logn')

p = pd.concat([uc, un], axis=1).dropna()
p = p[np.isfinite(p).all(axis=1)]

# How many households are actually observed more than once?
counts = p.groupby('i').size()
print(counts.value_counts().sort_index())

# Keep only households seen more than once -- singletons contribute nothing
# to a within estimator, and quietly inflate the apparent sample size.
p = p[counts.reindex(p.index.get_level_values('i')).to_numpy() > 1]

# Time dummies, then demean everything within household.
After demeaning
# there is no constant to include.
T = pd.get_dummies(p.index.get_level_values('t'), prefix='t',
```

```
                       drop_first=True, dtype=float)
T.index = p.index
q = pd.concat([p, T], axis=1)
qd = q - q.groupby(level='i').transform('mean')

res = sm.OLS(qd.logc, qd.drop(columns='logc')).fit(
    cov_type='cluster', cov_kwds={'groups': qd.index.get_level_values('i')}
print(res.summary().tables[1])
```

Compare the coefficient on `logn` here with the cross-sectional one from session 3. If they differ, the cross-sectional estimate was contaminated by whatever it is about large households that also predicts consumption.

4.11 The same panel, in welfare units

Everything above treats $\log c$ as the outcome. Session 3 argued that $w = -\log\lambda$ is the better welfare measure, because prices and household composition are swept into the good-time effects and the Barten scales rather than left in the number. Here is the same regression on w, so you can see how much it matters.

```
from pathlib import Path
import lsms_library as ll
import cfe
from cfe import Regression
import numpy as np, pandas as pd

def uganda_cfe():
    # The estimated CFE system for Uganda, as a cfe.Regression.
The object
    # carries beta, gamma, w = -log lambda, and predicted expenditures, so
    # nothing downstream has to refit.  Uses a copy staged on the hub if
    # there is one, else a copy you estimated earlier, else estimates it
    # from scratch (about forty seconds) and caches the result.
So this
    # cell is self-contained: no other notebook need have been run first.
    staged = Path('/srv/data/hhsurveys/uganda.rgsn')
    mine   = Path.home() / '.cache' / 'hhsurveys' / 'uganda.rgsn'
    for c in (staged, mine):
        if c.exists():
            return cfe.read_pickle(str(c))
```

```
    uga = ll.Country('Uganda')
    x = uga.food_expenditures().squeeze()
    agg = (uga.categorical_mapping['harmonize_food']
              .set_index('Preferred Label')['Aggregate Label'].to_dict())
    x = x.rename(index=agg, level='j')
    x = x.groupby(x.index.names).sum()

    y = np.log(x.replace(0, np.nan).dropna()).groupby(['i', 't', 'j']).sum(
    y = pd.concat({1: y}, names=['m']).reorder_levels(['i', 't', 'm', 'j'])

    d0 = uga.household_characteristics()
    d = d0.assign(
        Girls=d0[[f'F {a}' for a in ['00-03', '04-08', '09-13', '14-18']]].
        Boys =d0[[f'M {a}' for a in ['00-03', '04-08', '09-13', '14-18']]].
        Women=d0[[f'F {a}' for a in ['19-30', '31-50', '51+']]].sum(axis=1)
        Men  =d0[[f'M {a}' for a in ['19-30', '31-50', '51+']]].sum(axis=1)
    )[['Girls', 'Boys', 'Women', 'Men', 'log HSize']].dropna(how='any')
    d = d.groupby(['i', 't']).first()
    d = pd.concat({1: d}, names=['m']).reorder_levels(['i', 't', 'm']).sort

    r = Regression(y=y, d=d)
    r.get_beta(); r.get_w(); r.predicted_expenditures()
    mine.parent.mkdir(parents=True, exist_ok=True)
    r.to_pickle(str(mine))
    return r

r = uganda_cfe()
w = r.get_w()
w.groupby('t').agg(['size', 'mean', 'std']).round(3)

# The same within estimator, in welfare units.
# w arrives indexed (i, t, m) while p is (t, i).  Line them up explicitly:
# concat on a differently-ordered MultiIndex does not raise, it just fails
# to align -- the same trap that emptied the pseudo-panel above.
wi = w.droplevel('m') if 'm' in (w.index.names or []) else w
wi = wi.rename('w').reorder_levels(['t', 'i']).sort_index()

q2 = pd.concat([wi, p.logn], axis=1).dropna()
assert len(q2) > 0.5 * min(len(wi), len(p)), "w and p failed to align"
```

```
c2 = q2.groupby('i').size()
q2 = q2[c2.reindex(q2.index.get_level_values('i')).to_numpy() > 1]

T2 = pd.get_dummies(q2.index.get_level_values('t'), prefix='t',
                    drop_first=True, dtype=float)
T2.index = q2.index
Q2 = pd.concat([q2, T2], axis=1)
Q2d = Q2 - Q2.groupby(level='i').transform('mean')

res_w = sm.OLS(Q2d.w, Q2d.drop(columns='w')).fit(
    cov_type='cluster', cov_kwds={'groups': Q2d.index.get_level_values('i')
print(res_w.summary().tables[1])
```

Three comparisons to make, and the third is the one to remember.

The coefficient on `logn`. In $\log c$ it is about $+0.34$: a bigger household simply spends more. In w it is about -0.03 — small and *negative*. Household size has already been absorbed into the Barten scales, so what is left is what size does to welfare net of need, and that is slightly adverse. The two numbers are not rival estimates of one parameter; they answer different questions.

The time effects. In $\log c$ they climb steeply and monotonically, from 0.40 to 1.20 across the panel. That is mostly inflation, since nothing deflated those expenditures. In w they sit within ± 0.15 and do not trend, because prices live in a_t^j and never entered w at all. No CPI was used to achieve that, and none was available.

Now look at 2009–10 in the w column. It is the only negative time effect in the panel. That is the 2008 food price crisis, and it is invisible in $\log c$, where 2009–10 is simply another step up the inflation ladder. Session 5 takes that observation and asks what it does to the poverty rate.

4.12 Exercises

1. Rebuild the Ghana pseudo-panel with ten-year cohorts. How much do the lifecycle profiles change? Which conclusion is robust?

2. Implement the Verbeek–Nijman errors-in-variables correction using the within-cell variances, and report the corrected β.

3. In Uganda, quantify attrition: what fraction of the 2009–10 households appear in 2011–12? Are the leavers different in 2009–10 consumption from the stayers? What does that do to your fixed-effects estimate?

5 Capstone and Frontiers

Where the methods break down, and then your own analysis.

5.1 Reading

5.1.1 Reading

- Ligon, "Household Welfare from Disaggregate Demands" (working paper)
- Ligon (2026), "Risk sharing tests and covariate shocks: Drought, Floods, and Pests in Uganda," *AER*, forthcoming
- Townsend (1994), "Risk and insurance in village India," *Econometrica*

5.2 Two measures of poverty, and they disagree

Take Uganda through the 2008 food price crisis, when cereal prices more than doubled. Anchor a headcount at 58% in 2005–06 and carry the *same* threshold forward.

- on $w = -\log\lambda$: poverty **rises** to about 63% by 2009–10
- on nominal per-capita expenditure: the median rises about 32% over the same interval

Deflate by anything less than 32% inflation and you conclude that welfare improved. The two measures disagree in **sign**, not in magnitude.

5.3 Two measures of poverty, and they disagree

This is the reason session 3 spent time on rank, and it is worth stating the mechanism rather than leaving it as a surprising number.

Deflating nominal expenditure by a price index answers the question "how much more of the 2005 bundle could this household buy?" When relative prices move — cereals doubling while other prices do not — that question stops tracking welfare, because the household does not want the 2005 bundle any more, and because the bundle it does want has become more expensive in a way no single scalar records. With rank at least three, three indices would be needed; with one you get an answer whose error is correlated with exactly the price movement you are trying to measure.

The λ estimate never forms an index. It reads welfare off the composition of expenditure directly, and the composition is precisely what responds to relative prices. That is why it can move in the opposite direction, and why, when the two disagree, the burden of proof is not on it.

5.4 Full risk sharing

If a group of households shares risk fully, they equate marginal utilities of expenditure up to fixed Pareto weights:

$$\lambda_{it} = \frac{\mu_t}{\theta_i} \quad \Longleftrightarrow \quad w_{it} = \log \theta_i - \log \mu_t.$$

So under full insurance w_{it} is a household effect plus a time effect, and **nothing else**. The test writes itself:

$$w_{it} = \alpha_i + \delta_t + \gamma' S_{it} + \varepsilon_{it},$$

with S_{it} household-level shocks. Full risk sharing implies $\gamma = 0$.

5.5 Why covariate shocks break the test

The test above has a hole, and it is exactly where the policy interest is.

- δ_t absorbs **everything** common to the group in period t
- a drought that hits the whole region *is* common to the group
- so the covariate component of the shock is differenced away by the very time effect that makes the test valid

Consequence: the standard test has power only against **idiosyncratic** shocks. Failing to reject tells you nothing about insurance against drought, flood, or pests — the risks that actually threaten a rural economy, and the ones a village cannot self-insure.

5.6 Why covariate shocks break the test

There is a general lesson here that outlives this application. A fixed effect is not free: it removes variation, and you have to ask what was in the variation it removed. Session 4 made the point in passing — household effects absorb time-invariant selection and nothing else — but here the cost is sharper, because the absorbed variation is not a nuisance. It is the treatment.

The way out is not econometric cleverness so much as data: identify who was exposed to the covariate shock and who was not, using something outside the survey's own time dimension — rainfall, remote-sensed vegetation, the spatial footprint of a flood — and then the exposure varies within a period and survives the time effect. That is what the paper does, and it is the reason it needs the LSMS-ISA plot data alongside the panel.

5.7 The sign flip, computed

```
# Show tracebacks without the library's internal frames: the line that
# failed, and why.  Change Plain to Verbose if you ever want the rest.
%xmode Plain

# The first call that builds a table may print "DVC unavailable ...
# falling back to manual aggregation".  That is about how the library
# fetches its raw files, not about your data; the numbers are the same,
# and it does not recur once the table is built.

# These notebooks are read two ways: on your own screen, and projected onto
# a wall in a lit room where thin lines and pale markers are simply not
# there.  Those want different plots, so the difference is a switch rather
# than a compromise that suits neither.  Leave it False; the lecturer flips
# it.  Everything below the 'if' is about the wall, not about your laptop.
PROJECTOR = False

import matplotlib as mpl
mpl.rcParams.update({'figure.dpi': 110, 'font.size': 11,
                     'lines.linewidth': 1.8})
if PROJECTOR:
    mpl.rcParams.update({
        'figure.figsize': (7, 4.5),
        'font.size': 15, 'axes.labelsize': 15, 'axes.titlesize': 17,
        'xtick.labelsize': 13, 'ytick.labelsize': 13, 'legend.fontsize': 13
        'lines.linewidth': 3.0, 'lines.markersize': 9, 'axes.linewidth': 1.
    })

from pathlib import Path
import lsms_library as ll
import cfe
from cfe import Regression
```

```
import numpy as np, pandas as pd

def uganda_cfe():
    # The estimated CFE system for Uganda, as a cfe.Regression.
The object
    # carries beta, gamma, w = -log lambda, and predicted expenditures, so
    # nothing downstream has to refit.  Uses a copy staged on the hub if
    # there is one, else a copy you estimated earlier, else estimates it
    # from scratch (about forty seconds) and caches the result.
So this
    # cell is self-contained: no other notebook need have been run first.
    staged = Path('/srv/data/hhsurveys/uganda.rgsn')
    mine   = Path.home() / '.cache' / 'hhsurveys' / 'uganda.rgsn'
    for c in (staged, mine):
        if c.exists():
            return cfe.read_pickle(str(c))

    uga = ll.Country('Uganda')
    x = uga.food_expenditures().squeeze()
    agg = (uga.categorical_mapping['harmonize_food']
             .set_index('Preferred Label')['Aggregate Label'].to_dict())
    x = x.rename(index=agg, level='j')
    x = x.groupby(x.index.names).sum()

    y = np.log(x.replace(0, np.nan).dropna()).groupby(['i', 't', 'j']).sum(
    y = pd.concat({1: y}, names=['m']).reorder_levels(['i', 't', 'm', 'j'])

    d0 = uga.household_characteristics()
    d = d0.assign(
        Girls=d0[[f'F {a}' for a in ['00-03', '04-08', '09-13', '14-18']]].
        Boys =d0[[f'M {a}' for a in ['00-03', '04-08', '09-13', '14-18']]].
        Women=d0[[f'F {a}' for a in ['19-30', '31-50', '51+']]].sum(axis=1)
        Men  =d0[[f'M {a}' for a in ['19-30', '31-50', '51+']]].sum(axis=1)
    )[['Girls', 'Boys', 'Women', 'Men', 'log HSize']].dropna(how='any')
    d = d.groupby(['i', 't']).first()
    d = pd.concat({1: d}, names=['m']).reorder_levels(['i', 't', 'm']).sort

    r = Regression(y=y, d=d)
    r.get_beta(); r.get_w(); r.predicted_expenditures()
    mine.parent.mkdir(parents=True, exist_ok=True)
```

```
    r.to_pickle(str(mine))
    return r

r = uganda_cfe()
w = r.get_w()

waves = sorted(w.index.get_level_values('t').unique())
base = waves[0]

# Anchor at 58% in the base wave, then carry the SAME welfare threshold
# forward.  This is the whole exercise: one line, one threshold, no index.
z = w.xs(base, level='t').quantile(0.58)
hc = w.groupby('t').apply(lambda s: float((s <= z).mean()))
print(f"headcount on w, anchored at 58% in {base}:")
print(hc.round(3).to_string())

# The conventional measure, undeflated: nominal per-capita expenditure.
uga = ll.Country('Uganda')
xt = uga.food_expenditures().squeeze().groupby(['i', 't']).sum()
n = np.exp(uga.household_characteristics()['log HSize']).groupby(['i', 't']
pc = (xt / n.reindex(xt.index)).dropna()

med = pc.groupby('t').median()
print("median nominal per-capita food expenditure:")
print(med.round(0).to_string())
print(f"\nchange {waves[0]} -> {waves[1]}: "
      f"{100 * (med.loc[waves[1]] / med.loc[waves[0]] - 1):+.0f}%")
print(f"headcount on w, same interval: "
      f"{hc.loc[waves[0]]:.3f} -> {hc.loc[waves[1]]:.3f}")
```

One measure says welfare fell; the other says spending rose by a third. Reconciling them requires a deflator larger than the measured inflation over the period, which is to say: the conventional answer depends entirely on a number the survey does not contain.

5.8 A risk-sharing test

```
import statsmodels.api as sm

sh = uga.shocks()
print(sh.index.get_level_values('Shock').unique().tolist())
```

```python
# One indicator per shock type, per household-wave.
S = (sh.assign(hit=1.0).hit
        .groupby(['i', 't', 'Shock']).max()
        .unstack('Shock').fillna(0.0))

wi = w.droplevel('m') if 'm' in (w.index.names or []) else w
df = pd.concat([wi.rename('w'), S], axis=1).dropna(subset=['w'])
df[S.columns] = df[S.columns].fillna(0.0)

# Keep the shocks that are common enough to estimate.
keep = [c for c in S.columns if df[c].mean() > 0.02]
print(f"{len(keep)} shock types with >2% incidence")

D = pd.get_dummies(df.index.get_level_values('t'), prefix='t',
                   drop_first=True, dtype=float)
D.index = df.index
X = pd.concat([df[keep], D], axis=1)
Z = pd.concat([df.w, X], axis=1)
Zd = Z - Z.groupby(level='i').transform('mean')      # household effects

res = sm.OLS(Zd.w, Zd.drop(columns='w')).fit(
    cov_type='cluster', cov_kwds={'groups': Zd.index.get_level_values('i')}
print(res.summary().tables[1])
```

Read the shock coefficients, not the time dummies. Under full risk sharing every one of them is zero. Where they are not, that household bore the shock itself.

And then read what is missing. A drought that struck the whole country in a given wave contributes nothing to these estimates: the time dummy took it. Whatever this regression says about insurance, it says it only about the shocks that hit some households and not others in the same period.

5.9 Three more places it breaks

- **Short panels.** $y_{it} = \rho y_{i,t-1} + \alpha_i + \epsilon_{it}$ cannot be estimated by within: demeaning correlates the lagged dependent variable with the demeaned error (Nickell bias, order $1/T$). At $T = 3$ the bias in $\hat{\rho}$ can exceed 30%, downward. Arellano–Bond instruments in differences, and is weak precisely when ρ is near one.

- **Measurement error.** Recall error grows with the window and the item count, and it is not classical. Differencing removes signal and keeps noise, so fixed effects can make things worse, not better.

- **Endogeneity.** Land, labour supply, migration, schooling, adoption — all chosen. Fixed effects handle time-invariant selection only. Panel data does not solve endogeneity; it solves one special case of it.

5.10 Where it breaks

It is worth ending on this note because the temptation, having spent four sessions learning to handle these data carefully, is to over-claim with them. The instinct runs the wrong way: careful handling of the design makes the descriptive estimates trustworthy, and does nothing whatever for the causal ones.

The honest position is that household surveys are the best instrument we have for saying who is poor, where, by how much, and how that has changed — and that these are important questions which most of the profession's methodology is not aimed at. A well-constructed poverty profile with defensible standard errors is a more useful contribution than a badly-identified regression, and it is also harder to do. Session 5's purpose is to leave you willing to do the first and reluctant to do the second.

5.11 The capstone

Pick a country — `lsms_library.countries()` lists them — and produce, in about ninety minutes:

1. one paragraph on the design: frame, stages, strata, weights, structure
2. a consumption aggregate, or a defence of using the one provided
3. one weighted, correctly-clustered estimate you are willing to defend
4. one plot
5. a notebook that runs from a clean kernel, top to bottom

Five minutes each to present. The audience's job is to ask what the design lets you claim.

5.12 What counts as reproducible

- runs top to bottom in a fresh kernel, in order
- no manual steps, no "then I edited the CSV"
- data pulled by code, not sitting in a folder
- versions recorded — `lsms_library.__version__`, `pandas.__version__`
- someone else's machine, not just yours

This is not bureaucracy. It is the only way you will be able to answer a referee two years from now.

5.13 Setting up your capstone

```
import lsms_library as ll

# Everything the library knows about
ll.countries()

# Which tables exist where?  `coverage` is a country-by-table matrix.
# refresh='coverage' reads the declarations directly, so it does not need a
# prebuilt snapshot on disk.
cov = ll.coverage(refresh='coverage')
cov.head(20)

# Pick one and look before you leap
c = ll.Country('Uganda')          # <- change me
print(c.waves)
print(c.data_scheme)
```

5.14 A reproducibility check

Put this at the top of the notebook you present.

```
import sys, platform
from importlib.metadata import version
import lsms_library, pandas, numpy, statsmodels

print(f"python       {sys.version.split()[0]}  ({platform.system()})")
```

```python
print(f"lsms_library {lsms_library.__version__}")
print(f"CFEDemands   {version('CFEDemands')}")
print(f"pandas       {pandas.__version__}")
print(f"numpy        {numpy.__version__}")
print(f"statsmodels  {statsmodels.__version__}")
```

If you have edited a country's configuration or scripts, the parquet cache will be stale, and there is no automatic staleness check:

```python
# Force a rebuild for one country (slow; only when you have changed a confi
# !LSMS_NO_CACHE=1 python -c "import lsms_library as ll; ll.Country('Uganda
```

5.15 Exercises

The capstone is the exercise. Two suggestions if you finish early:

1. Take your estimate from (3) and compute it a second way — a different aggregate, a different scale, a different clustering level. Report the range. That range, not the standard error, is your real uncertainty.

2. Hand your notebook to the person next to you and have them run it from a clean kernel. Fix whatever breaks. That is the exercise.